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(54) MULTIPORT AIRFLOW CONTROL ASSEMBLY FOR AN AIR CUSHION

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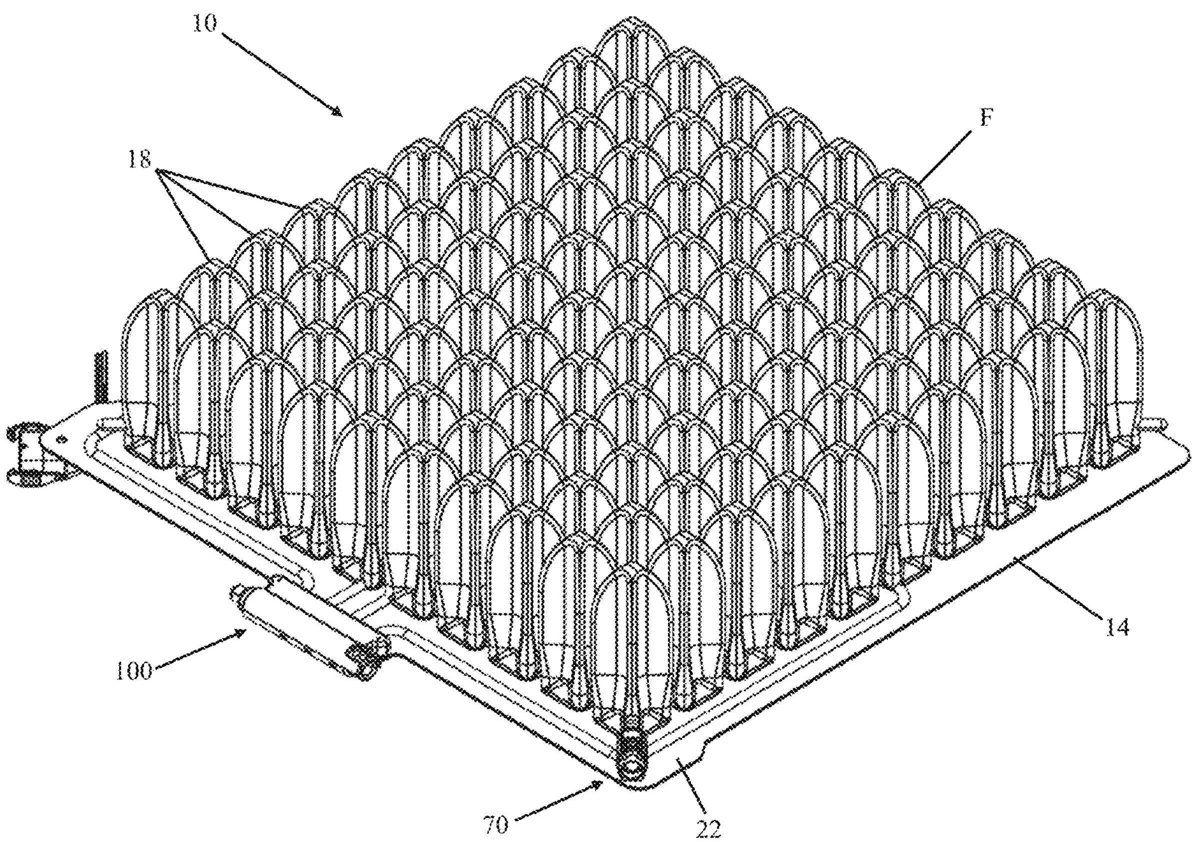
(57) **ABSTRACT**

A multiport airflow control manifold assembly includes a manifold assembly defining a first air channel, a second air channel, and a third air channel. A first first port and a first second port are fluidly connected to the first air channel. A second first port and a second second port are fluidly connected to the second air channel. A third first port and a third second port are fluidly connected to the third air channel. A seal member is fluidly connected to the manifold assembly. The seal member is configured to move between a plurality of seal configuration. In a first seal configuration, the seal member is configured to fluidly isolate the first air channel, the second air channel, and the third air channel. In a second seal configuration, the seal member is configured to fluidly connect the first air channel, the second air channel, and the third air channel.



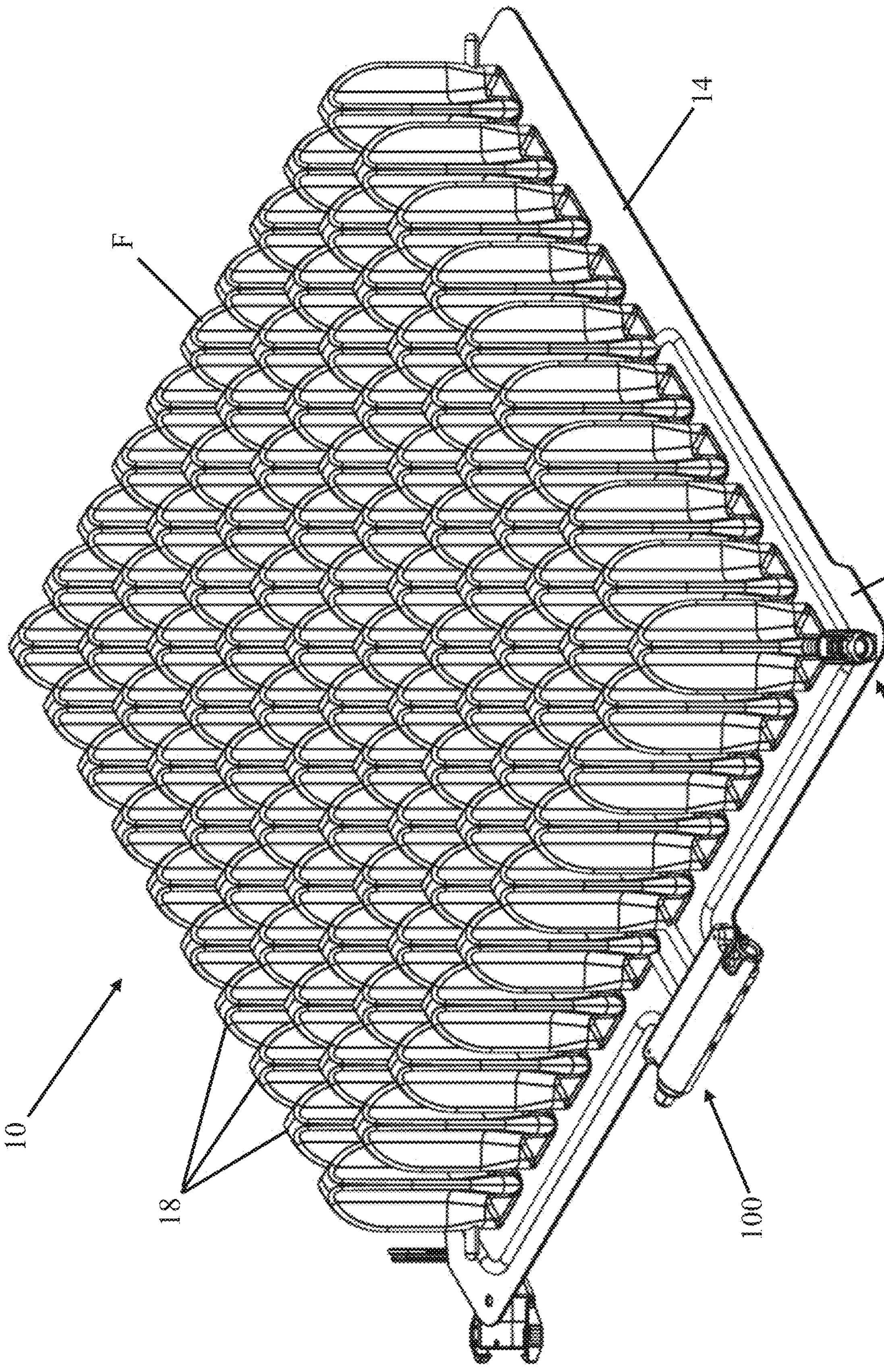


FIG. 1

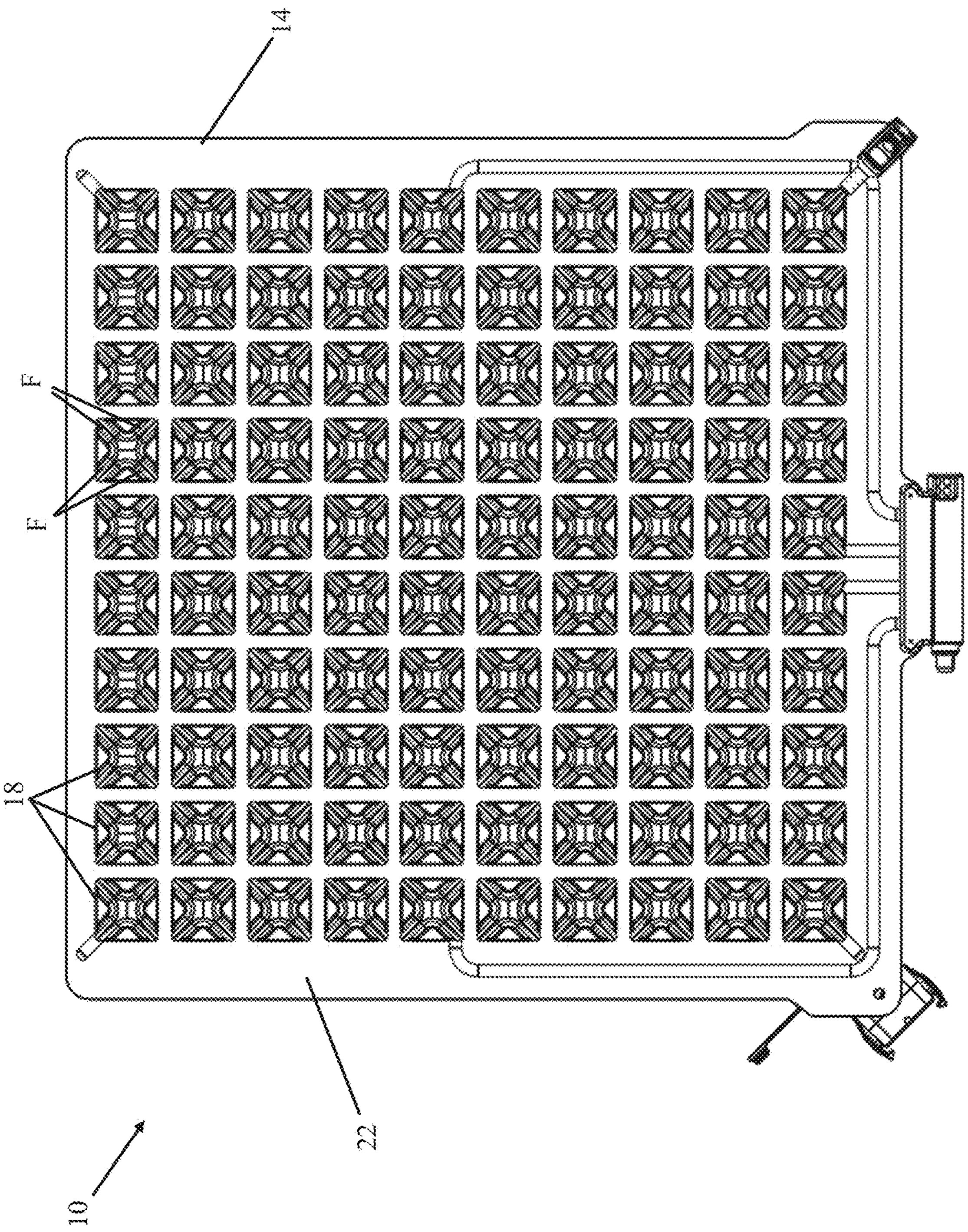


FIG. 2

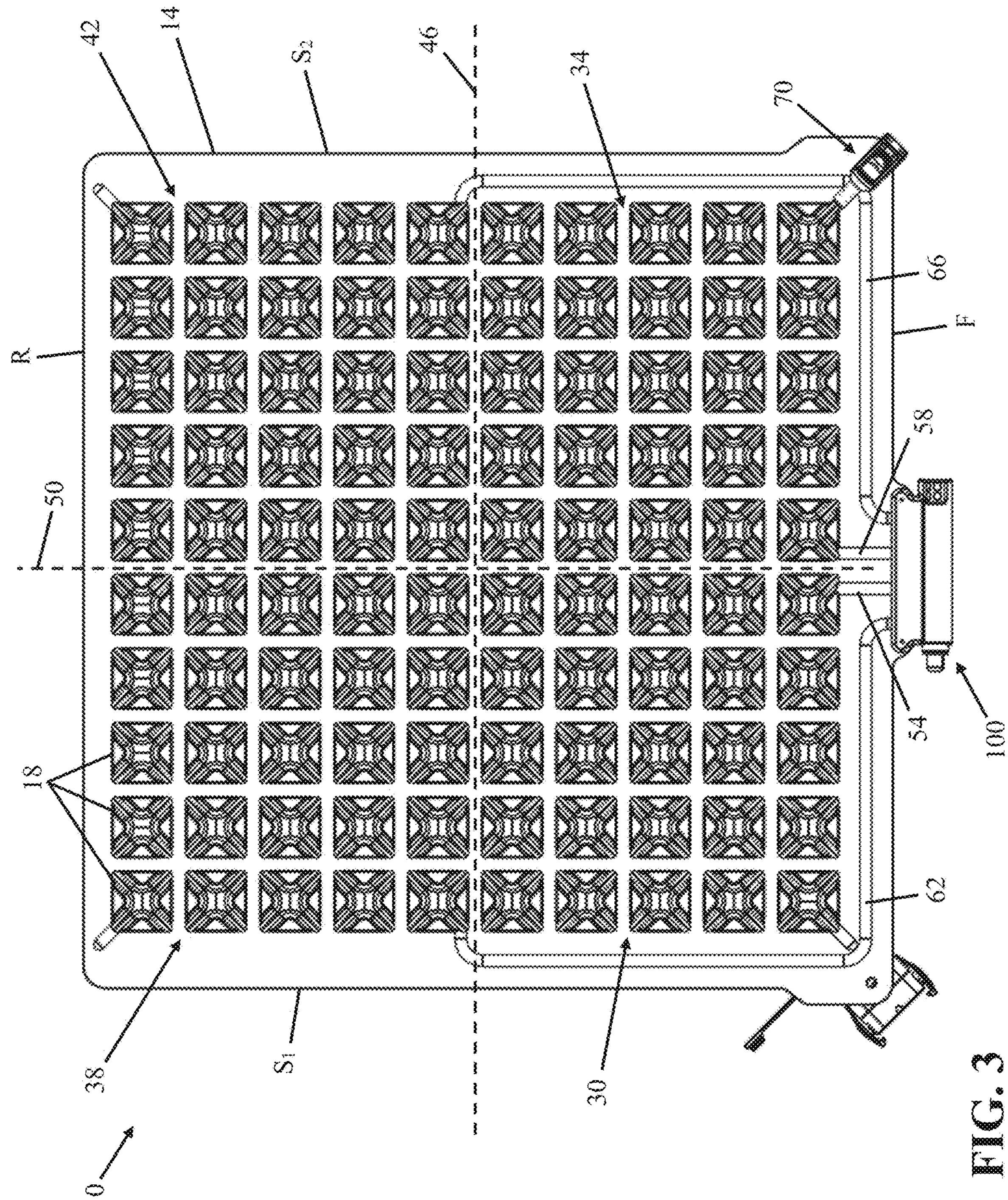


FIG. 3

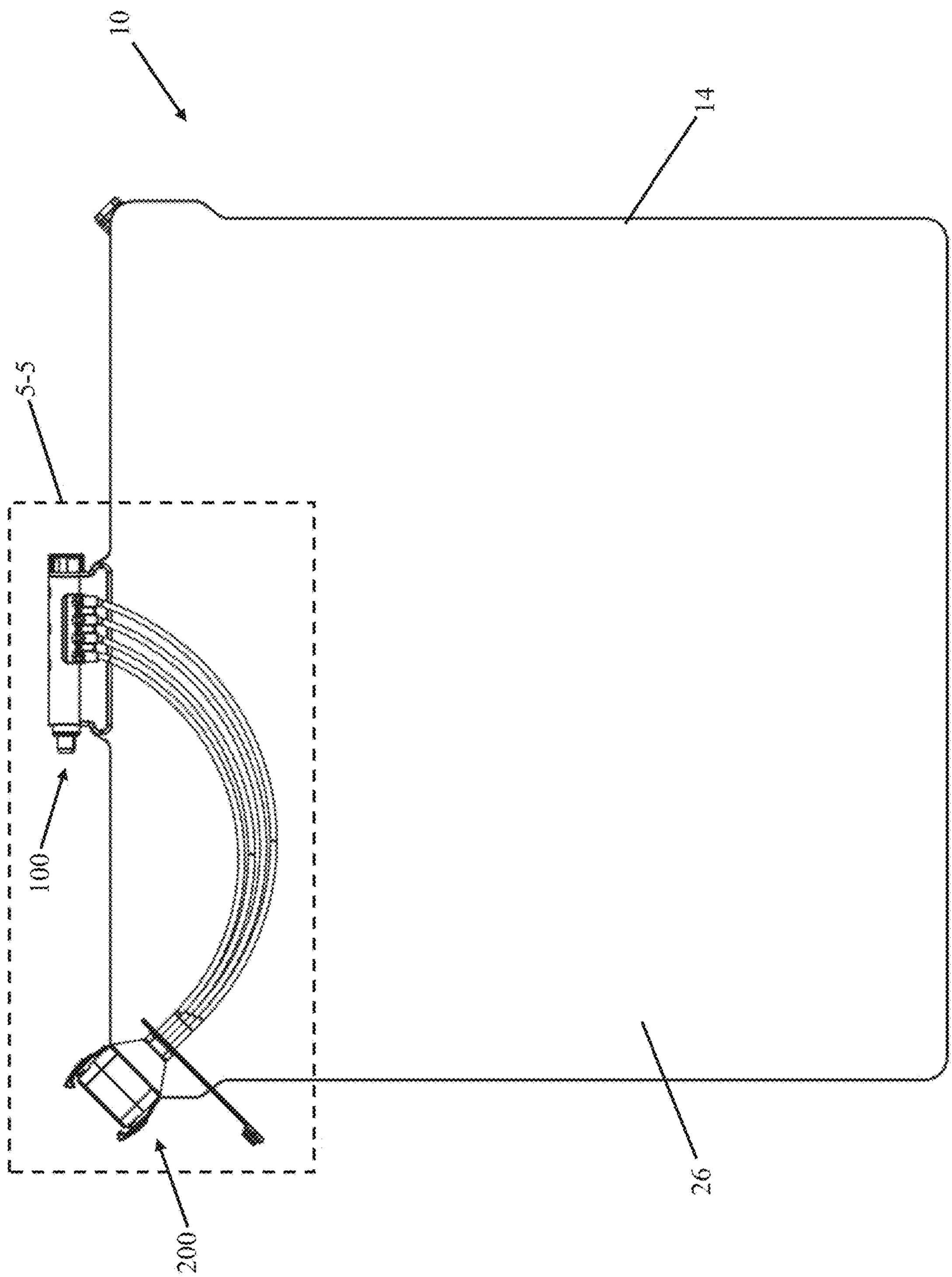


FIG. 4

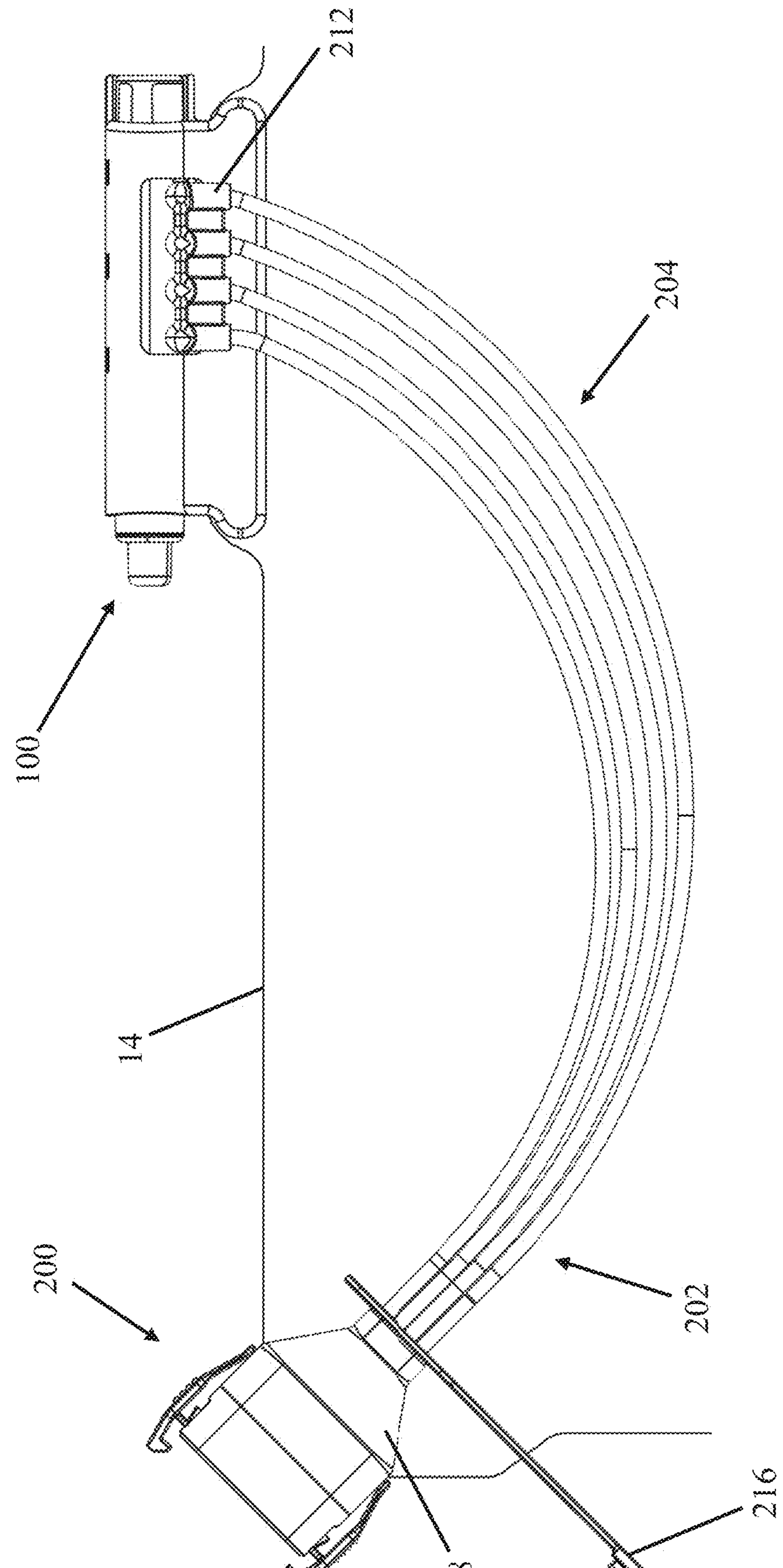


FIG. 5

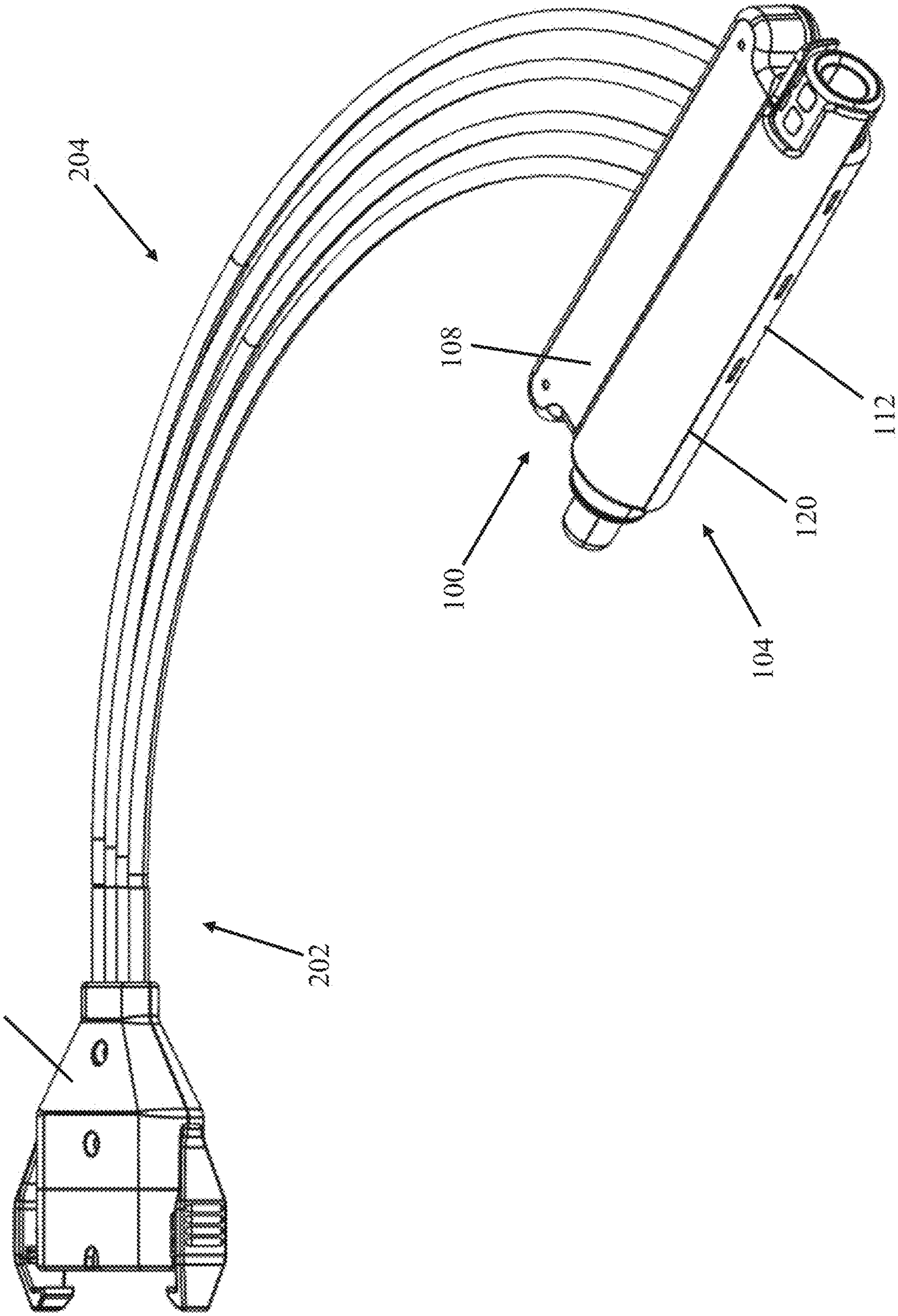


FIG. 6

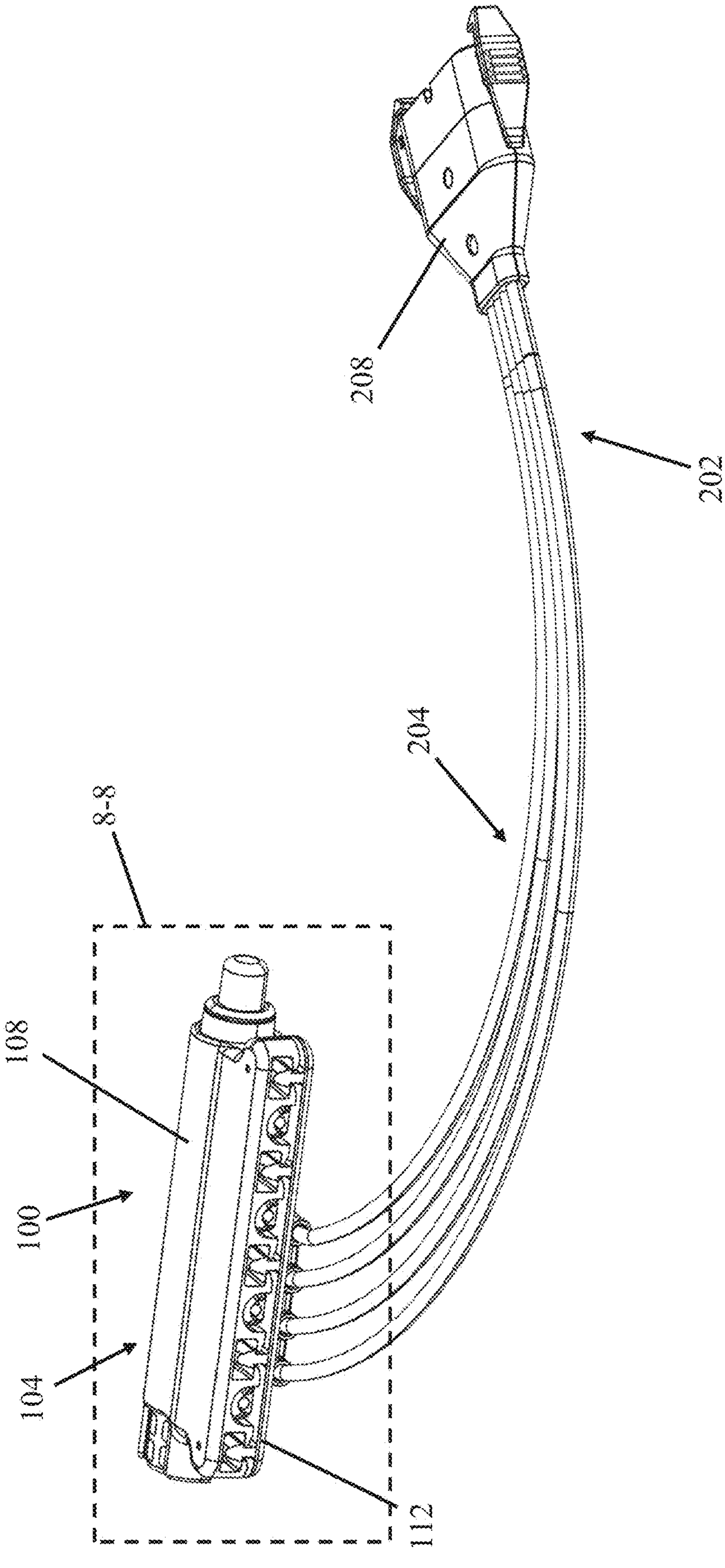


FIG. 7

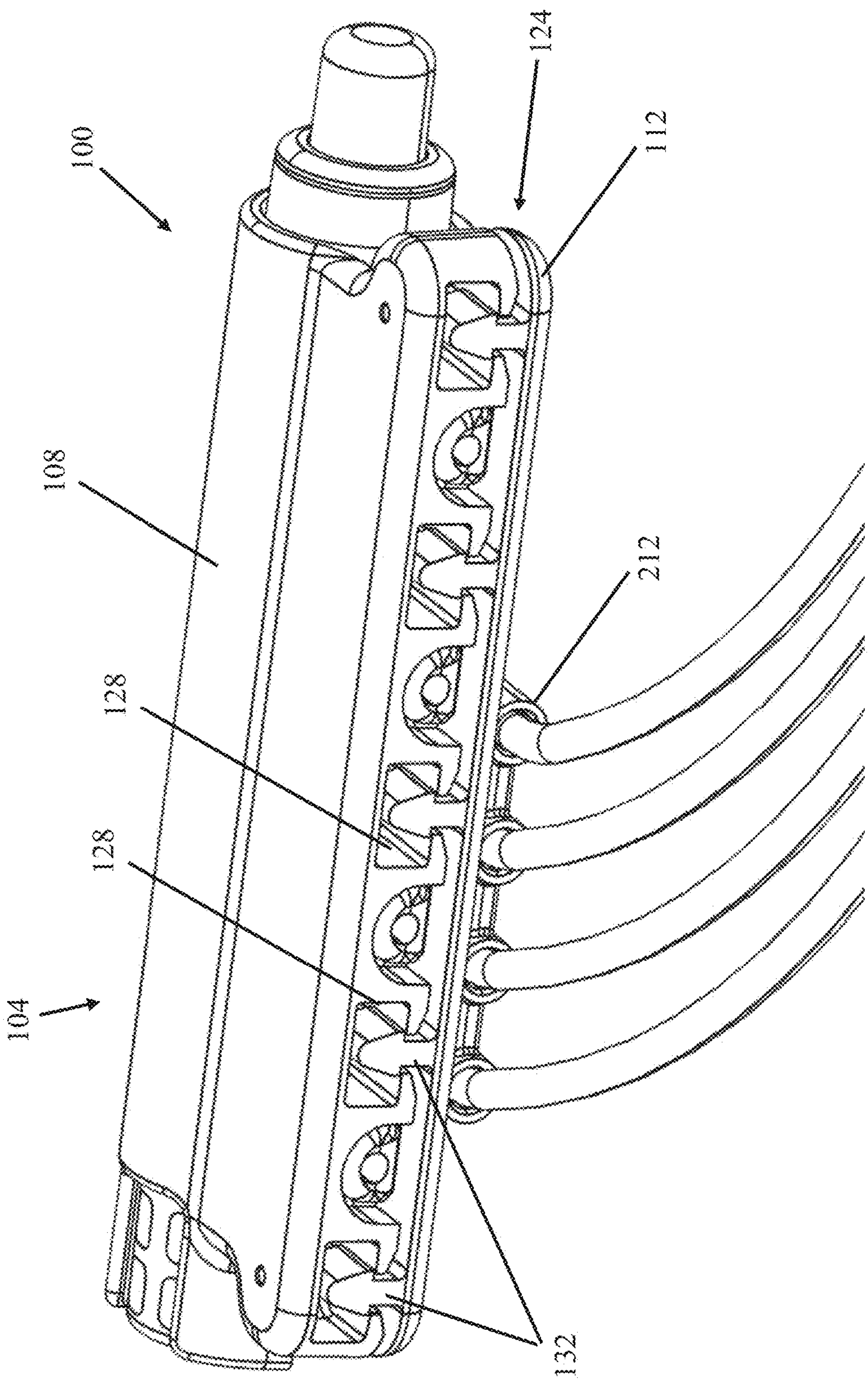


FIG. 8

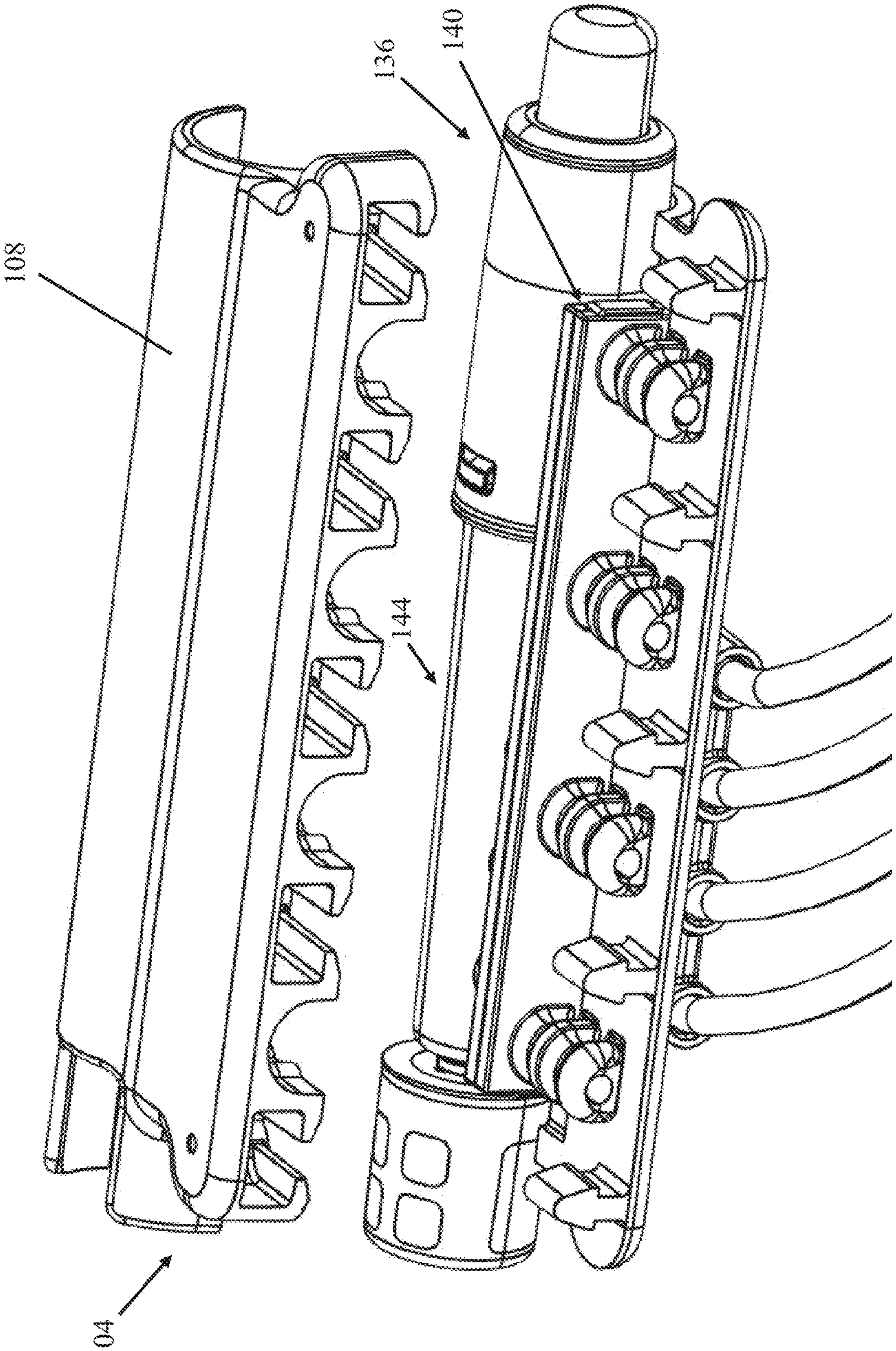


FIG. 9

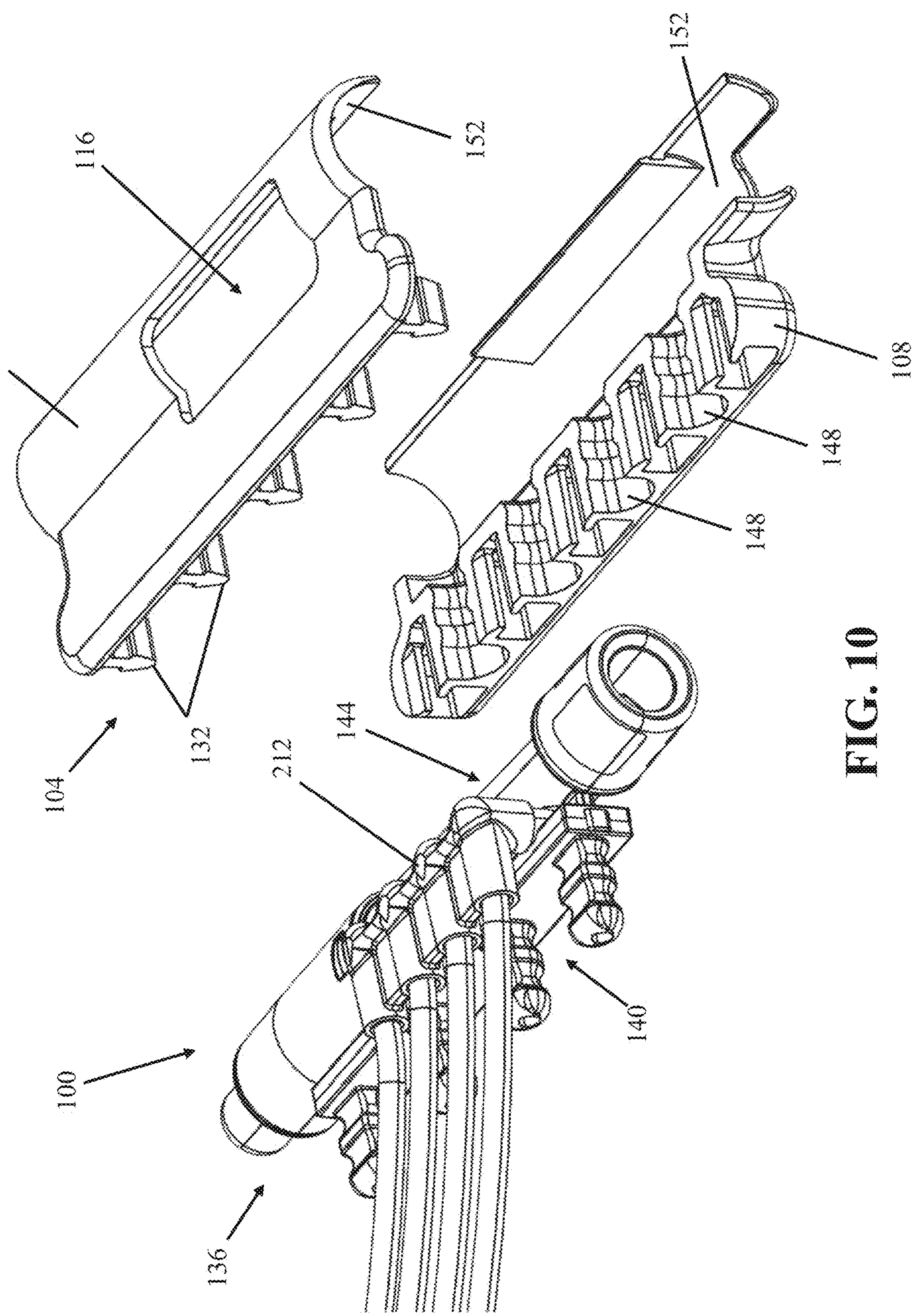


FIG. 10

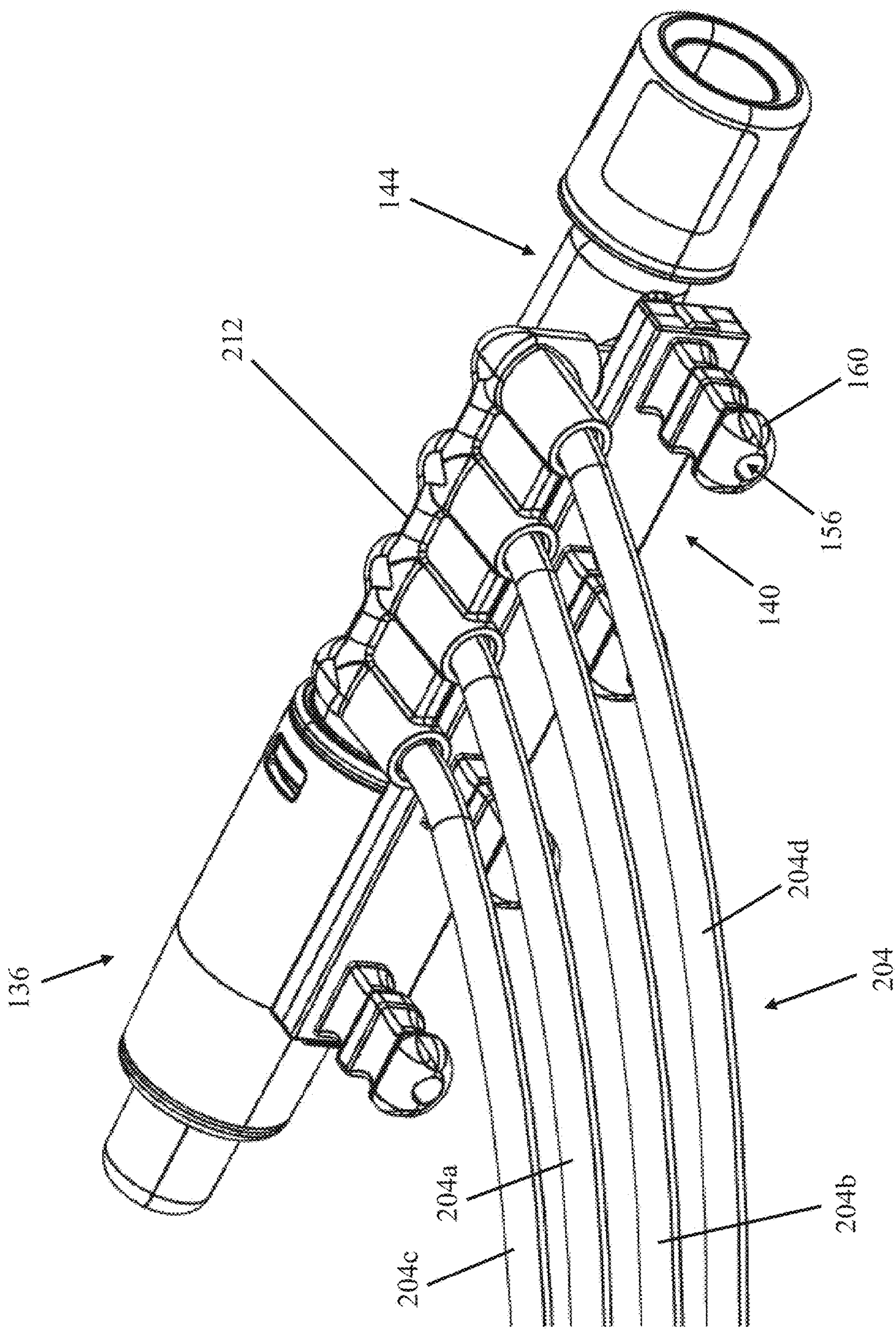


FIG. 11

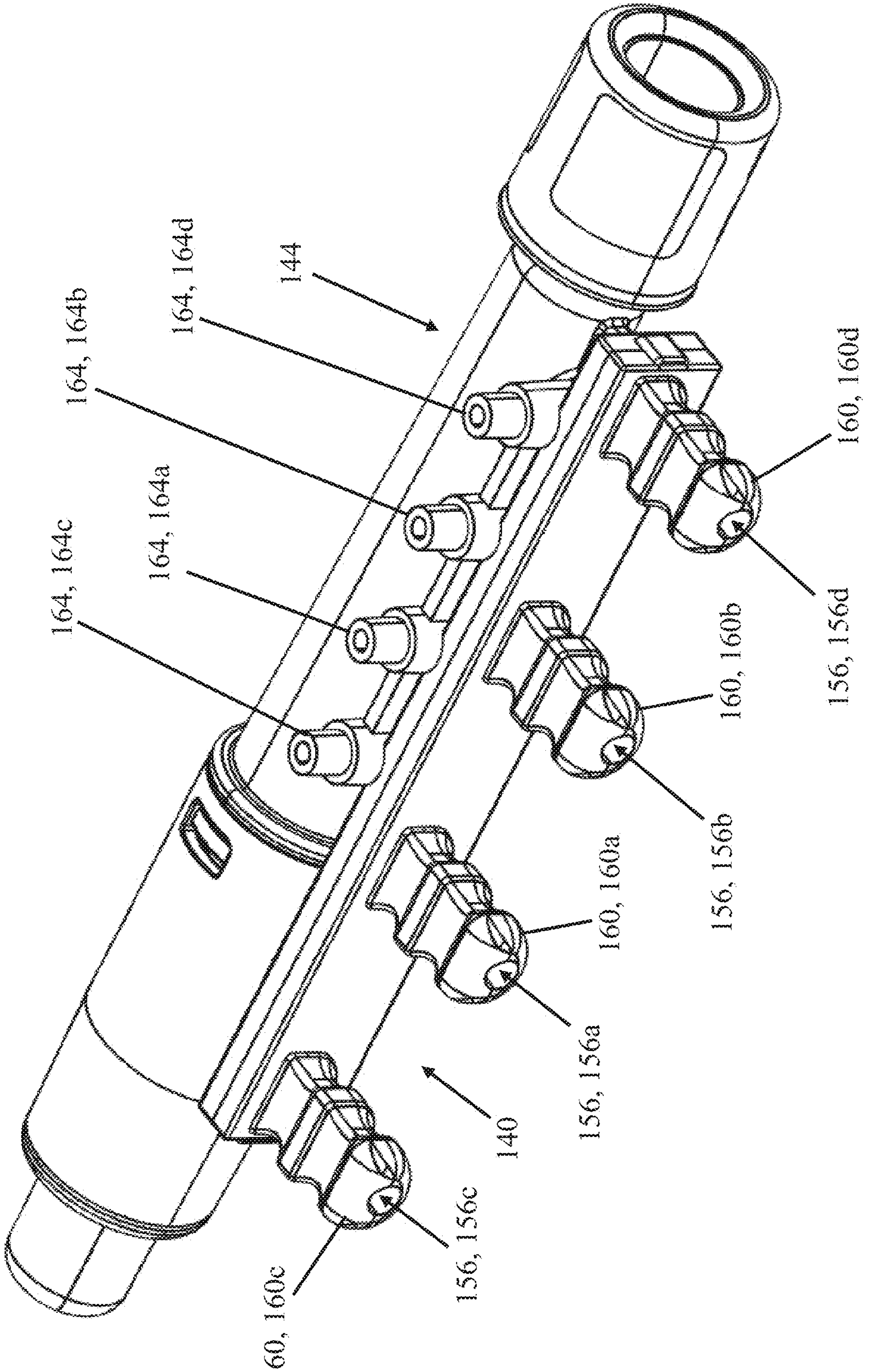

144
164, 164d
164, 164b
164, 164a
164, 164c
160, 160d
156, 156d
160, 160b
156, 156b
160, 160a
156, 156a
140
156, 156c
160, 160c

FIG. 12

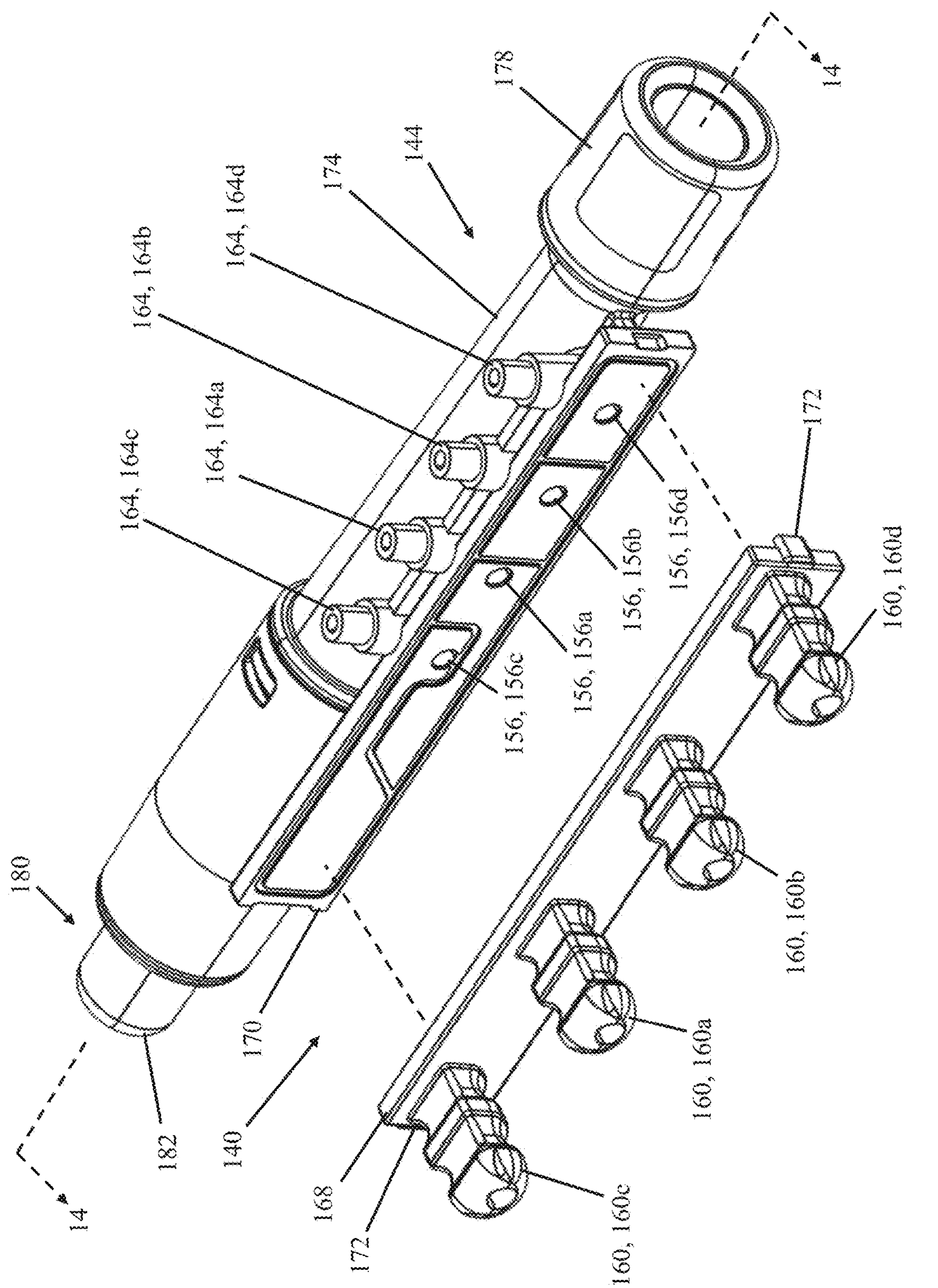

14
178
144
174
164, 164d
164, 164b
164, 164a
164, 164c
172
156, 156d
156, 156b
156, 156a
156, 156c
160, 160d
180
160, 160b
170
160, 160a
140
182
168
172
14
160, 160c

FIG. 13

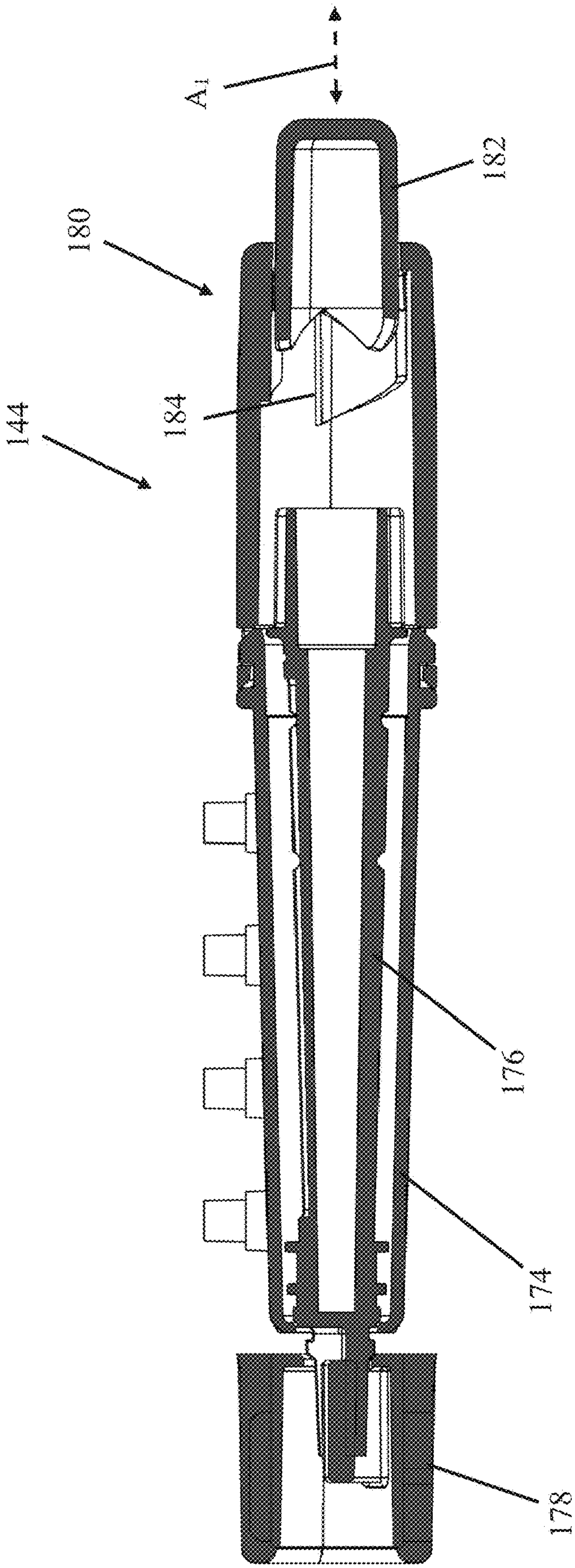


FIG. 14

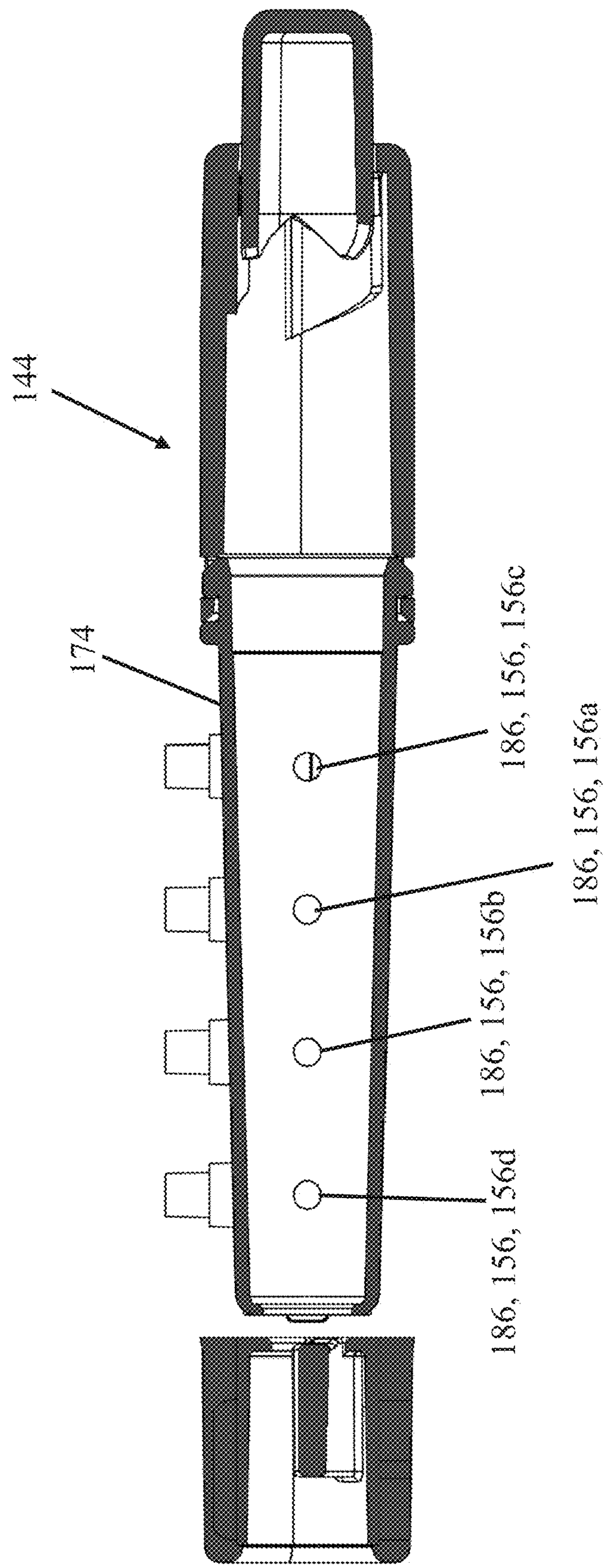


FIG. 15

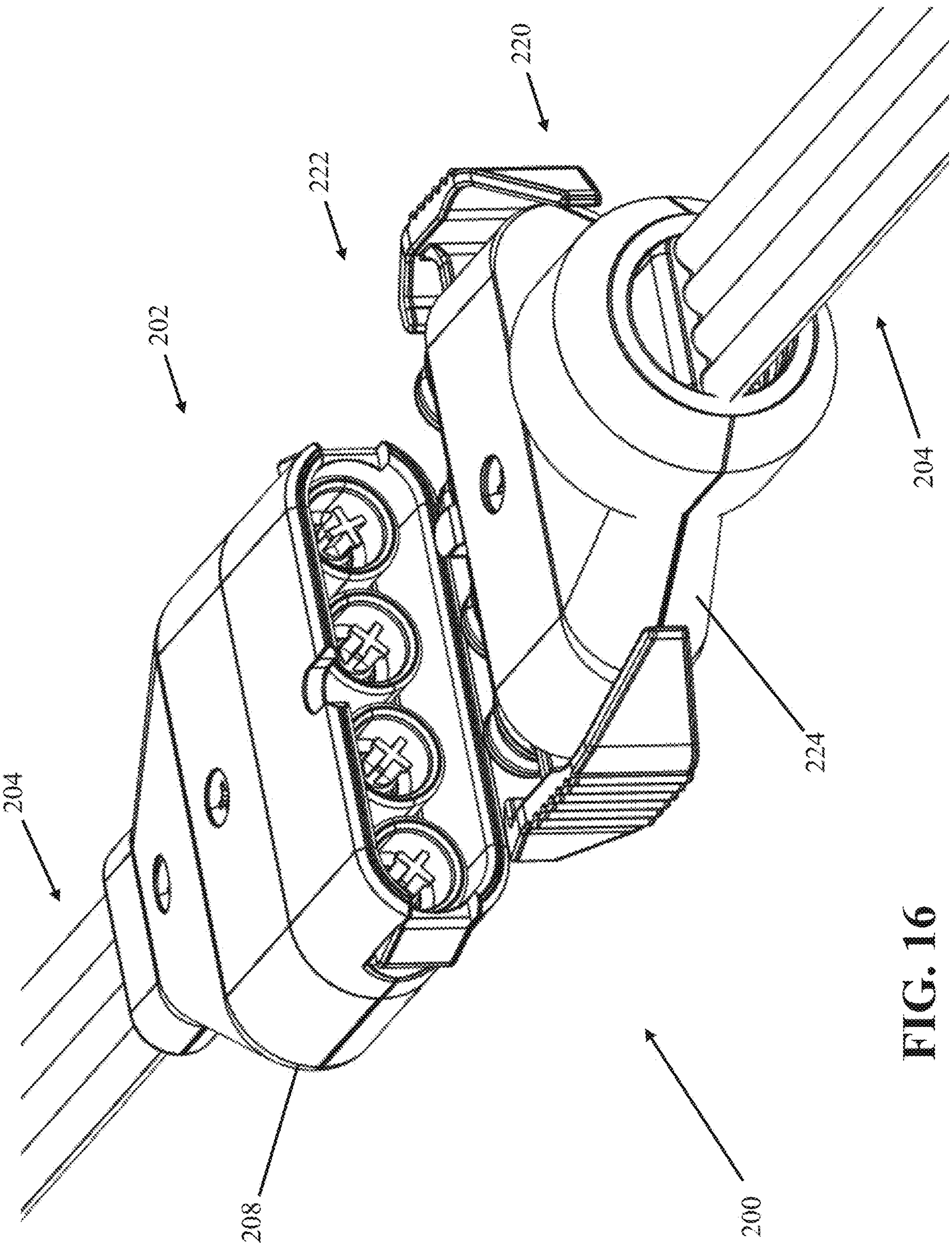


FIG. 16

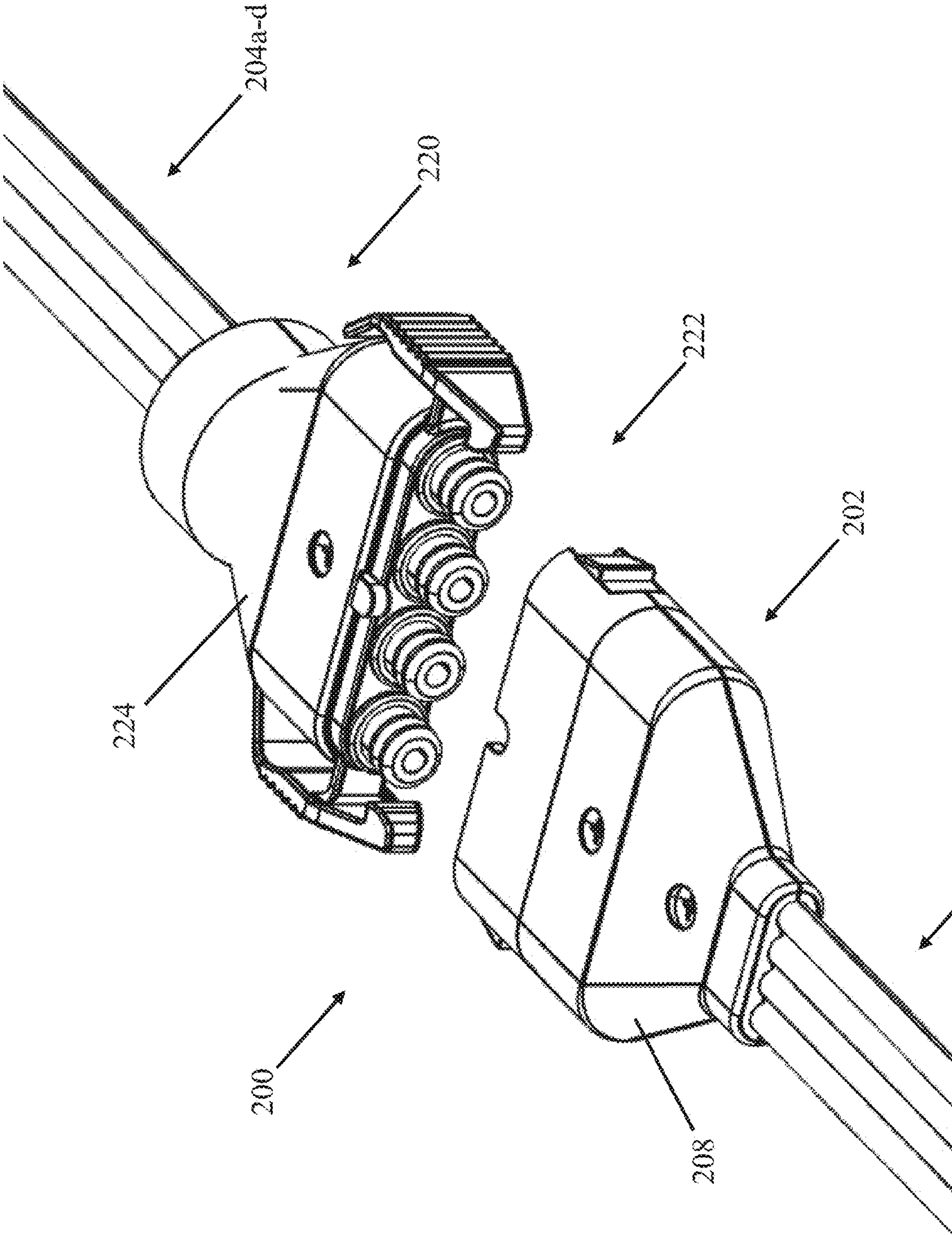
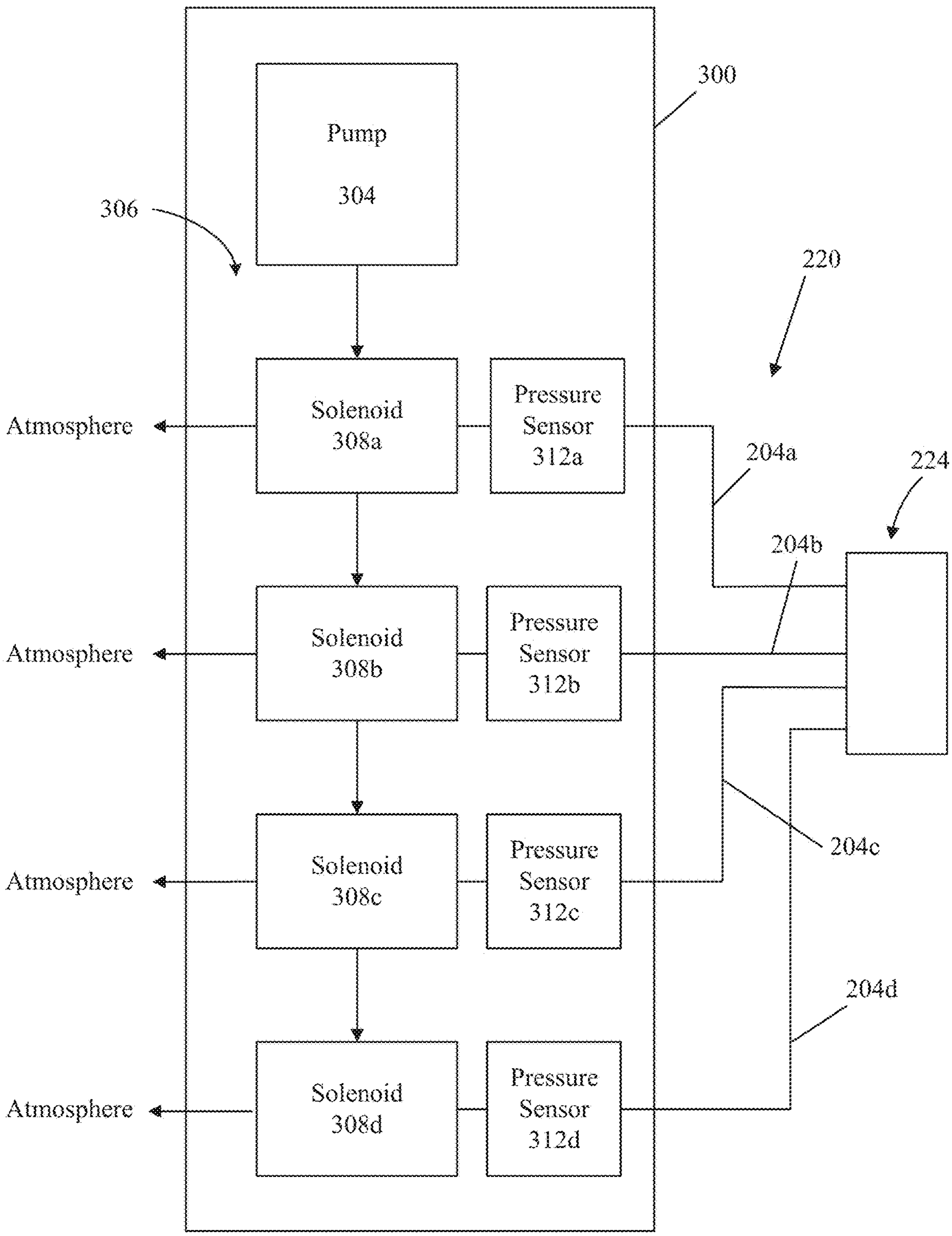


FIG. 17


300
Pump
304
306
220
Atmosphere
Solenoid
308a
Pressure
Sensor
312a
204a
224
204b
Atmosphere
Solenoid
308b
Pressure
Sensor
312b
204c
Atmosphere
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308c
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Sensor
312c
204d
Atmosphere
Solenoid
308d
Pressure
Sensor
312d

FIG. 18

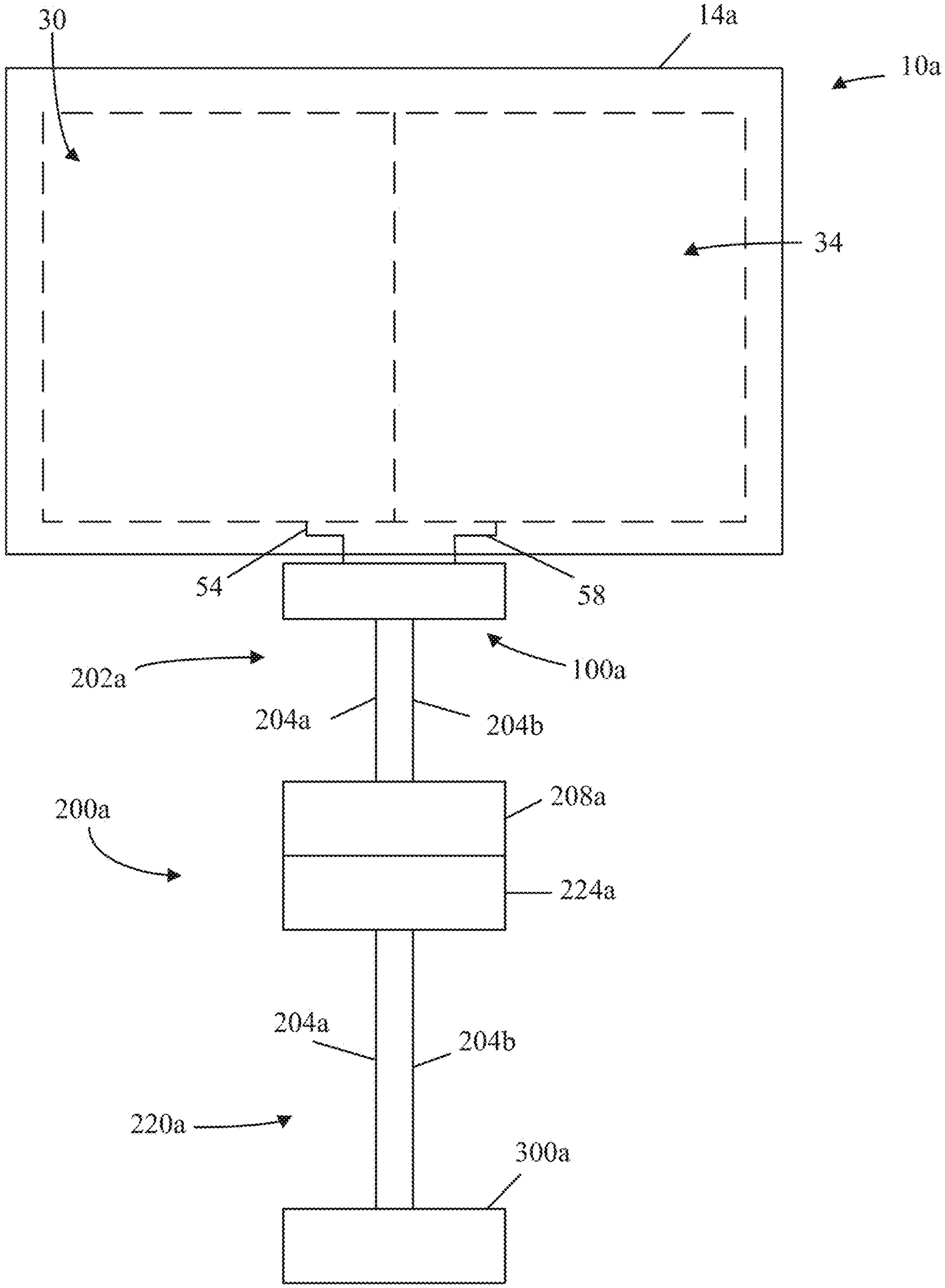
30
14a
10a
34
54
58
202a
100a
204a
204b
208a
200a
224a
204a
204b
220a
300a

FIG. 19

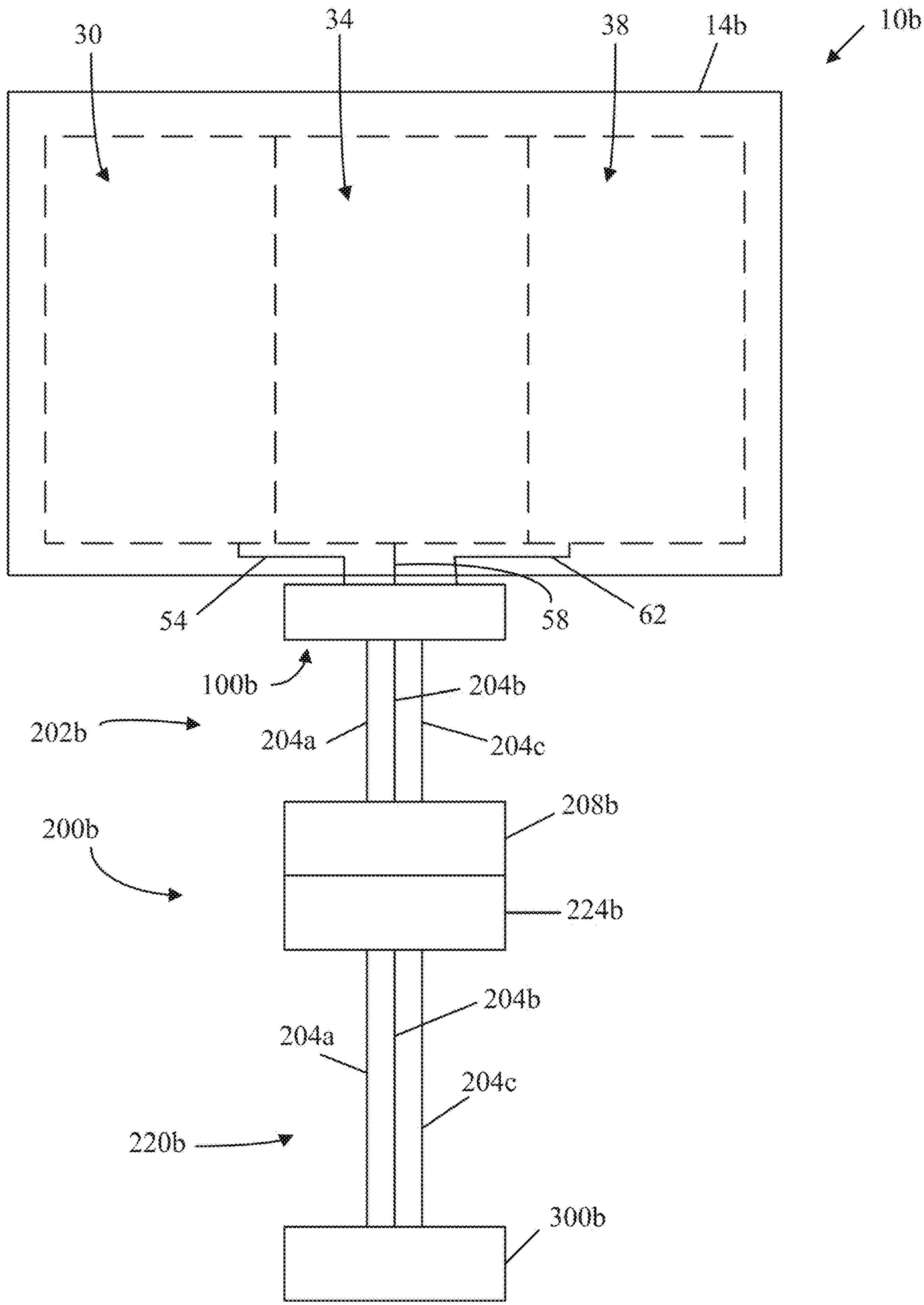
30
34
38
14b
10b
54
58
62
100b
204b
202b
204a
204c
208b
200b
224b
204b
204a
204c
220b
300b

FIG. 20

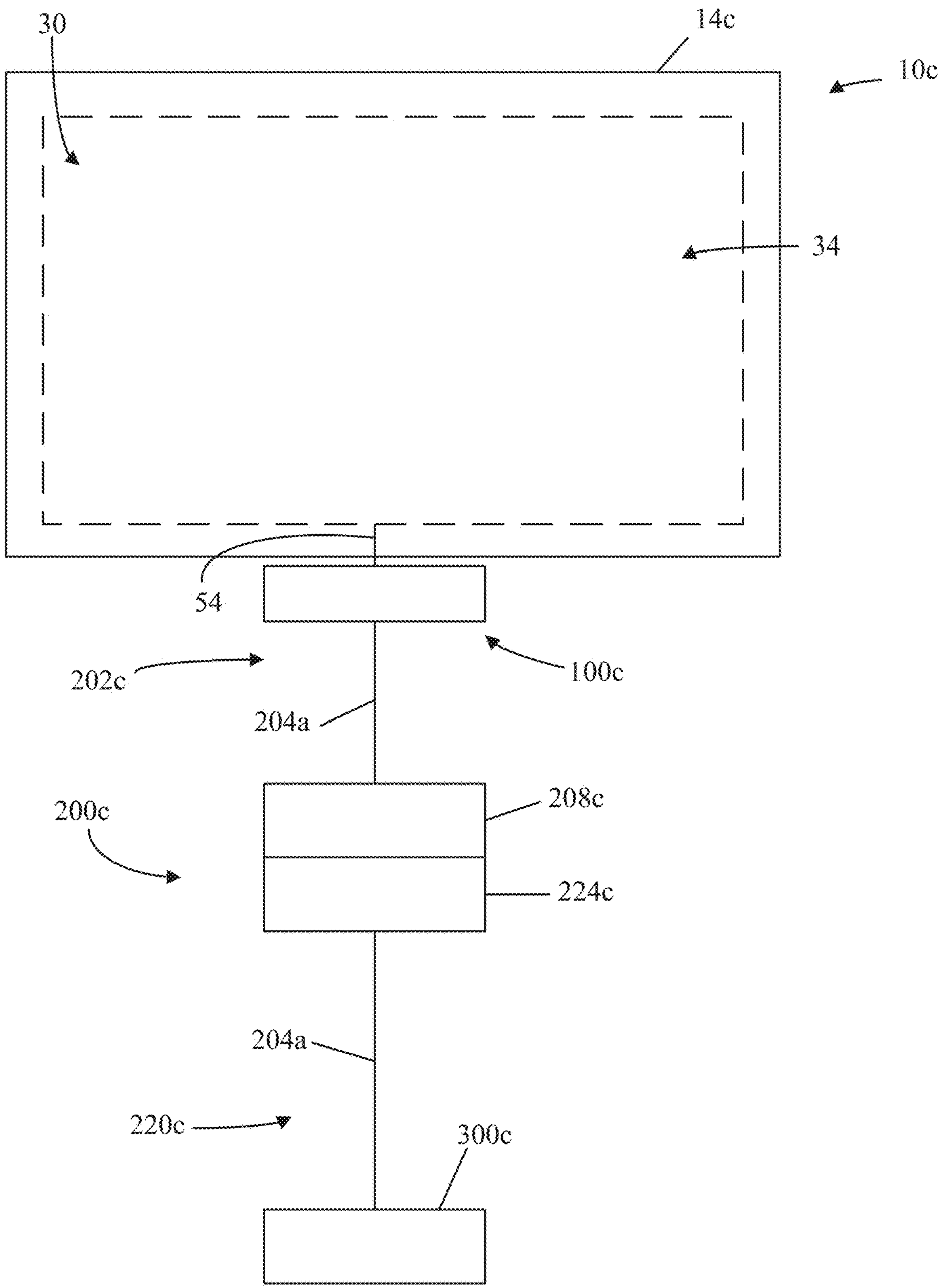

FIG. 21

MULTIPORT AIRFLOW CONTROL ASSEMBLY FOR AN AIR CUSHION

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates to an inflatable air cushion. More specifically, the present disclosure relates to a multiport airflow control assembly for use with an inflatable air cushion having a plurality of inflation zones. The airflow control assembly is configured to facilitate inflation, deflation, and to selectively fluidly connect different combinations of inflation zones to facilitate improved adjustability and customization of the inflatable air cushion for a user.

BACKGROUND

[0002] Individuals who are confined to wheelchairs are at higher risk of tissue breakdown and the development of pressure sores, which can be difficult to treat and/or cure. In certain circumstances, much of an individual's weight can concentrate in the region of the ischium, which includes the ischial tuberosity, or the bony prominence of the buttocks. Without regular movement, the flow of blood to the skin tissue in these regions can decrease, leading to the tissue damage and the development of pressure sores. Inflatable cellular air cushions are generally known to improve distribution of weight and thus provide protection from the occurrence of tissue damage and pressure sores. These cushions can include an array of air cells that project upwardly from a common base. Within the base the air cells are configured to communicate with each other, and thus, all exist at the same internal pressure. Hence, each air cell exerts essentially the same restoring force against the buttocks, irrespective of the extent to which it is deflected. U.S. Pat. No. 4,541,136 discloses such a cellular cushion currently manufactured and sold by Permobil, Inc. of Lebanon, Tennessee, USA for use on wheelchairs. The air cells can be separated into a plurality of air zones. What is needed is an improvement in a valve system for use with the air cushion that allows for remote inflation and deflation of one or more air zones, while also integrating functionality to selectively fluidly connect (or disconnect) one of more air zones for either separate or grouped remote inflation and deflation.

SUMMARY

[0003] In one example of an embodiment, a multiport airflow control manifold assembly includes a valve assembly including a manifold assembly defining a first air channel, a second air channel, and a third air channel. A first first port is fluidly connected to first air channel. A first second port is fluidly connected to the first air channel. A second first port is fluidly connected to second air channel. A second second port is fluidly connected to the second air channel. A third first port is fluidly connected to third air channel. A third second port is fluidly connected to the third air channel. An airflow control assembly is fluidly connected to the manifold assembly. The airflow control assembly includes a seal member configured to move between a first seal configuration and a second seal configuration. In the first seal configuration, the seal member is configured to fluidly isolate the first air channel, the second air channel, and the third air channel. In the second seal configuration, the seal member is configured to fluidly connect the first air channel, the second air channel, and the third air channel.

[0004] In another example of an embodiment, a multiport airflow control manifold assembly is configured to connect an air supply assembly to a cellular cushion, the multiport airflow control manifold assembly includes a manifold assembly defining a first air channel, a second air channel, and a third air channel. A first first port and a first second port are fluidly connected to the first air channel. A second first port and a second second port are fluidly connected to the second air channel. A third first port and a third second port fluidly are connected to the third air channel. An airflow control assembly is fluidly connected to the manifold assembly. The airflow control assembly includes a valve body fluidly connected to the first air channel, the second air channel, and the third air channel, and a seal member configured to move relative to the valve body between a first position and a second position. In the first position, the seal member is configured to fluidly isolate the first air channel, the second air channel, and the third air channel. In the second position, the seal member is configured to fluidly isolate the first air channel, and fluidly connect the second air channel and the third air channel.

[0005] Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a perspective view of an example of an embodiment of a cellular cushion and an embodiment of a multiport airflow control manifold assembly attached thereto.

[0007] FIG. **2** is a plan view of a top side of the cellular cushion of FIG. **1**.

[0008] FIG. **3** is a plan view of the cellular cushion of FIG. **2** illustrating a plurality of inflation zones that are separated by a plurality of imaginary axes.

[0009] FIG. **4** is a plan view of a bottom side of the cellular cushion of FIG. **1**.

[0010] FIG. **5** is an enhanced view of a multiport airflow control manifold assembly attached to the cellular cushion, as viewed from a bottom side of the cellular cushion and taken along line **5-5** of FIG. **4**.

[0011] FIG. **6** is a perspective view of the multiport airflow control manifold assembly and a first portion of an air supply assembly shown detached from the cellular cushion of FIG. **1**.

[0012] FIG. **7** is a perspective view of the multiport airflow control manifold assembly and first portion of the air supply assembly shown in FIG. **6**, illustrating a cushion engagement end of the multiport airflow control manifold assembly.

[0013] FIG. **8** is an enhanced view of the multiport airflow control manifold assembly, taken along line **8-8** of FIG. **7**.

[0014] FIG. **9** is a partially exploded perspective view of the multiport airflow control manifold assembly of FIG. **6**, illustrating a first housing member of a housing assembly detached from a second housing member to show a valve assembly received and retained therein.

[0015] FIG. **10** is a partially exploded view of the multiport airflow control manifold assembly of FIG. **6**, illustrating the valve assembly removed from the first and second housing members of the housing assembly.

[0016] FIG. **11** is a perspective view of the valve assembly of the multiport airflow control manifold assembly of FIG. **6** shown with the housing assembly removed for clarity.

[0017] FIG. **12** is a perspective view of the valve assembly of FIG. **11** shown with the air supply lines and manifold connector member removed for clarity.

[0018] FIG. **13** is a partially exploded perspective view of the valve assembly of FIG. **12** illustrating a manifold assembly with a first manifold member detached from a second manifold member.

[0019] FIG. **14** is a cross-sectional view of an airflow control assembly of the valve assembly of FIG. **12**, taken along line **14-14** of FIG. **13**.

[0020] FIG. **15** is a cross-sectional view of the airflow control assembly of FIG. **14** showing the seal member removed to illustrate apertures defined by a valve body to fluidly connect an associated air channel to the valve body.

[0021] FIG. **16** is a perspective view of a first portion and a second portion of the air supply assembly shown in a disengaged configuration and illustrating a first coupling member.

[0022] FIG. **17** is a perspective view of a first portion and a second portion of the air supply assembly shown in a disengaged configuration and illustrating a second coupling member.

[0023] FIG. **18** is a schematic diagram of an example of an embodiment of a remote air source assembly coupled to the second portion of the air supply assembly.

[0024] FIG. **19** is a schematic diagram of another example of an embodiment of a cellular cushion, a multiport airflow control manifold assembly, an air supply assembly, and a remote air source assembly depicting the cellular cushion with two inflation zones.

[0025] FIG. **20** is a schematic diagram of another example of an embodiment of a cellular cushion, a multiport airflow control manifold assembly, an air supply assembly, and a remote air source assembly depicting the cellular cushion with three inflation zones.

[0026] FIG. **21** is a schematic diagram of another example of an embodiment of a cellular cushion, a multiport airflow control manifold assembly, an air supply assembly, and a remote air source assembly depicting the cellular cushion with one inflation zone.

[0027] Before embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The disclosure is capable of supporting other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

[0028] The present disclosure is directed to an embodiment of a multiport airflow control manifold assembly **100** configured for operation with a cellular cushion **10**. The multiport airflow control manifold assembly **100** is configured to facilitate selective inflation and/or deflation of each of a plurality of inflation zones **30**, **34**, **38**, **42** of the cellular cushion **10**. The inflation and/or deflation can be in response to a detected or targeted pressure in the associated plurality of inflation zones **30**, **34**, **38**, **42**. In addition, the multiport airflow control manifold assembly **100** is configured to selectively fluidly connect and/or fluidly isolate different combinations of the plurality of inflation zones **30**, **34**, **38**, **42** of the cellular cushion **10**.

[0029] With reference now to the figures, FIG. **1** is a perspective view of an example of an embodiment of a cellular cushion **10** (also referred to as a cellular air cushion **10**). The cellular cushion **10** includes a base **14** and a plurality of air cells **18**. The base **14** is formed of an upper layer **22** and a lower layer **26** (shown in FIG. **4**). The upper layer **22** can be coupled to the lower layer **26** (or backing layer **26**) by a suitable adhesive. In the illustrated embodiment, the upper layer **22** can be formed of a first material while the lower layer **26** can be formed of a second, different material. In one example of an embodiment, the upper layer **22** can be formed of a flexible neoprene, while the lower layer **26** can be formed of polyurethane. In other examples of embodiments, the upper layer **22** and the lower layer **26** can be formed of any material (or combination of materials) suitable for operation of the cellular cushion **10** as described herein. An example of the cellular cushion **10** is disclosed in U.S. Pat. No. 4,541,136, the contents of which is hereby incorporated by reference in its entirety.

[0030] With reference to FIGS. **1-2**, the plurality of air cells **18** project away from the base **14**. The plurality of air cells **18** are molded into the upper layer **22**, and thus are defined by the upper layer **22**. In addition, the upper layer **22** interconnects the plurality of air cells **18**. Each of the plurality of air cells **18** include four fins F. In other examples of embodiments, each of the plurality of air cells **18** can have any suitable configuration, including but not limited to air cells **18** having any number of fins, any number of sides, or having no fins (e.g., cylindrical cells, cubical cells, rounded cells, etc.).

[0031] The plurality of air cells **18** are arranged on the upper layer **22** in a plurality of longitudinal and transverse rows. As such, each air cell **18** occupies both a longitudinal row and a transverse row. In other examples of embodiments, the plurality of air cells **18** can be arranged in any geometry suitable for providing support to a user. For example, the air cells **18** can be arranged in a semi-circular pattern, a circular pattern, or any other suitable arrangement or geometry of air cells **18** to facilitate support for a user.

[0032] With reference now to FIG. **3**, the cellular cushion **10** is arranged into a plurality of zones. More specifically, the air cells **18** of the cellular cushion **10** are arranged into a plurality of zones (also referred to as inflation zones or support zones). In the illustrated embodiment, the plurality of zones include four different inflation zones **30**, **34**, **38**, **42**. A first zone **30** is positioned adjacent to a second zone **34** along a first axis **46**. Stated another way, the first and second zones **30**, **34** are positioned side by side at a front F (or first end F) of the cellular cushion **10**. A third zone **38** is positioned adjacent to a fourth zone **42** along the first axis **46**. Stated another way, the third and fourth zones **38**, **42** are positioned side by side at a rear R (or second end R) of the cellular cushion **10**. In addition, the first zone **30** and the third zone **38** are positioned side by side along a second axis **50**. Stated another way, the first and third zones **30**, **38** are positioned side by side along a right side S_1 (or a first side S_1) of the cellular cushion **10**. The second zone **34** and the fourth zone **42** are also positioned side by side along the second axis **50**. Stated another way, the second and fourth zones **34**, **42** are positioned side by side along a left side S_2 (or a second side S_2) of the cellular cushion **10**. It should be appreciated that the sides (i.e., right and left sides) are described in relation to the user sitting on the cellular cushion **10**. When describing the sides in relation to viewing the cellular cushion **10** from the front F to the rear R, the first side S_1 can be referred to as a left side, and the second side S_2 can be referred to as a right side. In the illustrated embodiment, the first axis **46** is generally perpendicular to (or generally orthogonal to) the second axis **50**. In other embodiments of the cellular cushion **10**, the zones **30**, **34**, **38**, **42** of air cells **18** (shown in FIG. **1**) can be oriented in any suitable orientation or geometry relative to each other. It should be appreciated that each of the plurality of zones **30**, **34**, **38**, **42** is generally fluidly isolated from any other of the plurality of zones **30**, **34**, **38**, **42**. For example, during manufacturing, the upper and lower layers **22**, **26** can be adhesively fastened along the imaginary first and second axes **46**, **50** to define the plurality of fluidly isolated zones **30**, **34**, **38**, **42**. Thus, the air cells **18** positioned in each zone **30**, **34**, **38**, **42** can be fluidly connected, but the air cells **18** positioned in each zone **30**, **34**, **38**, **42** are fluidly isolated from air cells **18** in another zone **30**, **34**, **38**, **42**. Fluid connection between zones **30**, **34**, **38**, **42** can only be facilitated by the multiport airflow control manifold assembly **100**, which is discussed in further detail below.

[0033] The base **14** defines a plurality of fluid conduits **54**, **58**, **62**, **66** (also referred to as an air conduit **54**, **58**, **62**, **66**). Each conduit **54**, **58**, **62**, **66** fluidly connects each of the plurality of zones **30**, **34**, **38**, **42** to a multiport airflow control manifold assembly **100**. A first fluid conduit **54** fluidly connects the first zone **30** to the manifold assembly **100**. A second fluid conduit **58** fluidly connects the second zone **34** to the manifold assembly **100**. A third fluid conduit **62** fluidly connects the third zone **38** to the manifold assembly **100**. A fourth fluid conduit **66** fluidly connects the fourth zone **42** to the manifold assembly **100**. It should be appreciated that each conduit **54**, **58**, **62**, **66** can be formed by molding, vacuum forming, or any other suitable process for formation between the upper and lower layers **22**, **26** of the base **14** (shown in FIGS. **2** and **4**).

[0034] An air valve **70** (also referred to as an inflation-deflation valve **70**) is fluidly connected to the plurality of air cells **18**. In the illustrated embodiment, the air valve **70** is fluidly connected to one of the air zones, and more specifically the second zone **34**. In other examples of embodiments, the air valve **70** can be fluidly connected to the first zone **30**, the third zone **38**, or the fourth zone **42**. The air valve **70** is configured to facilitate manual inflation and/or deflation of the plurality of air cells **18** (or inflation and deflation of the plurality of zones **30**, **34**, **38**, **42**). For example, the air valve **70** can be configured to engage an air pump (not shown) to facilitate inflation. The air pump can be a hand pump, a manual pump, a motorized pump, or any other suitable pump that is configured to supply air to the cellular cushion **10**. The air pump (not shown) can provide a flow of air to one zone (for example, the second zone **34**, the first zone **30**, etc.). In the illustrated embodiment, air travels from the second zone **34** to the multiport airflow control manifold assembly **100** through the conduit **58**. The air is then distributed to the other zones **30**, **38**, **42** through the respective conduits **54**, **62**, **66** by the manifold assembly **100**. Similarly, the air valve **70** can be configured to deflate the plurality of cells **18** of the cellular cushion **10**. The air valve **70** can be opened to the atmosphere, facilitating a release of air within the plurality of cells **18**. More specifically, air can flow from the first, third, and fourth zones **30**, **38**, **42** to the second zone **34**. The air flows through the respective conduits **54**, **62**, **66** to the manifold assembly **100**, where it is directed through the conduit **58** to the second zone **34**, and then discharged from the cellular cushion **10** through the air valve **70**. It should be appreciated that the air valve **70** includes certain components and operates in a manner disclosed in U.S. Pat. No. 11,739,854, which is entitled "Valve Assembly for an Air Cushion" and assigned to Permobil, Inc., the entire contents of which is herein incorporated by reference in its entirety.

[0035] With reference now to FIGS. **3-4**, the multiport airflow control manifold assembly **100** is configured to selectively fasten to the base **14**. More specifically, the manifold assembly **100** is configured to fluidly engage each conduit **54**, **58**, **62**, **66**, while also fastening to the base **14**. The manifold assembly **100** is also configured to fluidly connect to a remote air source. The remote air source is configured to supply airflow to each of the respective plurality of zones **30**, **34**, **38**, **42**.

[0036] With reference now to FIGS. **4-5**, an air supply assembly **200** is configured to attach to the multiport airflow control manifold assembly **100**. The air supply assembly **200** is configured to connect the remote air source to the multiport airflow control manifold assembly **100**. With specific reference to FIG. **5**, a first portion **202** of the air supply assembly **200** is illustrated. The first portion **202** includes a plurality of air supply lines **204**. The air supply lines **204** are each connected at a first end to a coupling member **208**. The coupling member **208** is illustrated as a female coupling member **208**. The air supply lines **204** are each connected at a second end, opposite the first end, to a manifold connector member **212**. The connector member **212** is configured to attach each the plurality of air supply line **204** to the multiport airflow control manifold assembly **100**. In the illustrated embodiment, the connector member **212** is a ninety degree (90°) adapter configured to engage the multiport airflow control manifold assembly **100**. In this embodiment, one end of the connector member **212** is oriented orthogonal to the opposite end, such that the end in engagement with the multiport airflow control manifold assembly **100** is orthogonal (or perpendicular) to the opposite end in engagement with the plurality of air supply lines **204**. In other examples of embodiments, the connector member **212** can be oriented in any suitable or desired orientation or angle. A clip member **216** can also be provided to selectively fasten (or clip) the first portion **202** of the air supply assembly **200** to the base **14**. The clip member **216** can be provided to maintain a position of the first portion **202** relative to the base **14**. It should be appreciated that in other examples of embodiments the coupling member **208** can be a male coupling member **208**.

[0037] With reference now to FIGS. **6-8**, the multiport airflow control manifold assembly **100** and the first portion **202** of the air supply assembly **200** are shown detached from the base **14** of the cellular cushion **10**. The multiport airflow control manifold assembly **100** includes a housing assembly **104**. The housing assembly **104** includes a first housing member **108** and a second housing member **112**. The first housing member **108** (or upper housing member **108**) is configured to be viewable (or exposed) to a user in response to fastening the multiport airflow control manifold assembly **100** to the cellular cushion **10**. The second housing member **112** defines an aperture **116** (shown in FIG. **10**). The aperture **116** is configured to receive the connector member **212**. The first and second housing members **108**, **112** are configured to self-fasten together. For example, the first and second housing members **108**, **112** are configured to connect at a first end **120** (shown in FIG. **6**). The connection between the first and second housing members **108**, **112** can be formed by complimentary interlocking components or a living hinge. The first and second housing members **108**, **112** are configured to self-fasten at a second end **124** (shown in FIG. **8**). With reference to FIG. **8**, the first housing member **108** can define a plurality of first fasteners **128**, while the second housing member **112** can define plurality of second fasteners **132**. Each first fastener **128** is configured to engage a second fastener **132** to fasten the first and second housing members **108**, **112**, and defining the self-fastening end **124** of the multiport airflow control manifold assembly **100**. In the illustrated embodiment, the first fastener **128** is illustrated as an elongated channel, while the second fastener **132** is illustrated as a projection that is configured to be received and retained by the channel. In other examples of embodiments, any suitable fasteners **128**, **132** can be used to fasten (or self-fasten) the housing members **108**, **112** together. It should also be appreciated that the fastening end **124** is configured to engage and retain (or trap) a portion of the base **14** of the cellular cushion **10** between the housing members **108**, **112**. This facilitates fastening of the multiport airflow control manifold assembly **100** to the cellular cushion **10**.

[0038] With reference now to FIGS. **9-10**, the housing assembly **104** houses a valve assembly **136**. The valve assembly **136** includes a manifold assembly **140** and an airflow control assembly **144**. The housing assembly **104** is configured to receive and retain the valve assembly **136**. In addition, the housing assembly **104** can include a geometry that is complimentary to the valve assembly **136** to receive and retain the valve assembly **136**. For example, as shown in FIG. **10**, the housing assembly **104** includes a first portion that defines a plurality of channels **148** (or recesses **148**) that are configured to receive a portion of the manifold assembly **136**. As another example, and also shown in FIG. **10**, the housing assembly includes a second portion that defines an arcuate surface **152** that is configured to receive and retain the airflow control assembly **144**. In other examples of embodiment, the housing assembly **104** can include any suitable geometry that is complimentary or suitable to the valve assembly **136** to restrict undesirable movement of the valve assembly **136** relative to the housing assembly **104**, and further retain the valve assembly **136** in the housing assembly **104**.

[0039] FIGS. **11-13** illustrate the valve assembly **136** removed from the housing assembly **104**. The manifold assembly **140** includes at least one first air channel **156**. With specific reference to FIG. **12**, each air channel **156** is in fluid communication with a first port **160**, a second port **164**, and the airflow control assembly **144**. In the illustrated embodiment, the manifold assembly **140** includes a plurality of air channels **156**, and more specifically four air channels **156**. In other examples of embodiment, the valve assembly **136** includes a number of air channels **156** that corresponds to the number of inflation zones **30**, **34**, **38**, **42** of the cellular cushion **10**. For example, the valve assembly **136** can include one air channel **156**, two air channels **156**, three air channels **156**, four air channels **156**, or five or more air channels **156**.

[0040] With continued reference to FIG. **12**, each air channel **156** is in fluid communication with one first port **160** and one second port **164**. Each air channel **156** is also in fluid communication with the airflow control assembly **144**. For each air channel **156**, the first port **160** is downstream of the second port **164**, and the second port **164** is downstream of the airflow control assembly **144**. Stated another way, the second port **164** is fluidly connected to the air channel **156** upstream of the first port **160** and downstream of the airflow control assembly **144**. Stated yet another way, both the first and second ports **160**, **164** are in fluid communication with the air channel **156** downstream of the airflow control assembly **144**, with the first port **160** being downstream of the second port **164**.

[0041] With regard to the embodiment illustrated in FIG. **12**, the manifold assembly **140** includes a first air channel **156*a***, a second air channel **156*b***, a third air channel **156*c***, and a fourth air channel **156*d***. The first air channel **156*a*** is in fluid communication with a first first port **160*a*** and a first second port **164*a***. The first air channel **156*a*** is also in fluid communication with the airflow control assembly **144**. The second air channel **156*b*** is in fluid communication with a second first port **160*b*** and a second second port **164*b***. The second air channel **156*b*** is also in fluid communication with the airflow control assembly **144**. The third air channel **156*c*** is in fluid communication with a third first port **160*c*** and a third second port **164*c***. The third air channel **156*c*** is also in fluid communication with the airflow control assembly **144**. The fourth air channel **156*d*** is in fluid communication with a fourth first port **160*d*** and a fourth second port **164*d***. The fourth air channel **156*d*** is also in fluid communication with the airflow control assembly **144**.

[0042] Each respective air channel **156*a-d*** of the multiport airflow control manifold assembly **100** is in fluid communication with a respective inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**. To facilitate the fluid communication between the multiport airflow control manifold assembly **100** and each respective inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**, each first port **160** is configured to be in fluid communication with a respective inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**. In the illustrated embodiment, the first first port **160*a*** is in fluid communication with the first zone **30** of the cellular cushion **10** (shown in FIG. **3**). To facilitate the fluid communication, in the illustrated example, the first first port **160*a*** is in fluid connection with the first fluid conduit **54**. More specifically, the first fluid conduit **54** is configured to receive the first first port **160*a***.

[0043] The second first port **160*b*** is in fluid communication with the second zone **34** of the cellular cushion **10** (shown in FIG. **3**). To facilitate the fluid communication, in the illustrated example, the second first port **160*b*** is in fluid connection with the second fluid conduit **58**. More specifically, the second fluid conduit **58** is configured to receive the second first port **160*b***.

[0044] The third first port **160*c*** is in fluid communication with the third zone **38** of the cellular cushion **10** (shown in FIG. **3**). To facilitate the fluid communication, in the illustrated example, the third first port **160*c*** is in fluid connection with the third fluid conduit **62**. More specifically, the third fluid conduit **62** is configured to receive the third first port **160*c***.

[0045] The fourth first port **160*d*** is in fluid communication with the fourth zone **42** of the cellular cushion **10** (shown in FIG. **3**). To facilitate the fluid communication, in the illustrated example, the fourth first port **160*d*** is in fluid connection with the fourth fluid conduit **66**. More specifically, the fourth fluid conduit **66** is configured to receive the fourth first port **160***d*.

[0046] With reference to FIG. **13**, the manifold assembly **140** can include a first manifold member **168** and a second manifold member **170**. The first manifold member **168** includes the one or more first ports **160**, while the second manifold member **170** includes the one or more second ports **164**. The manifold members **168**, **170** are configured to fasten together, for example by a fastener **172** positioned on each end of the first manifold member **168**. The manifold members **168**, **170** cooperate to define the air channels **156**.

[0047] As discussed in additional detail below, a remote air source is in fluid communication with each second port **164** of the multiport airflow control manifold assembly **100**. Accordingly, the remote air source, which is configured to selectively supply air and/or withdraw air, is in fluid communication with each respective air channel **156***a-d*. To facilitate the fluid communication between the remote air source and the multiport airflow control manifold assembly **100**, the remote air source is fluidly connected to the multiport airflow control manifold assembly **100** by the air supply assembly **200**. More specifically, the air supply lines **204** are configured to fluidly connect the remote air source to the multiport airflow control manifold assembly **100**. With reference back to FIG. **11**, a plurality of the air supply lines **204** are connected to the connector member **212**, which is in turn connected to each of the second ports **164** (shown in FIG. **12**). In the illustrated example of embodiment, a first air supply line **204***a* is fluidly connected to the first second port **164***a*, a second air supply line **204***b* is fluidly connected to the second second port **164***b*, a third air supply line **204***c* is fluidly connected to the third second port **164***c*, and a fourth air supply line **204***d* is fluidly connected to the fourth second port **164***d*. In the illustrated embodiment, the connector member **212** connects each air supply line **204***a-d* to the respective second port **164***a-d*. However, in other embodiments, each air supply line **204***a-d* can be connected to each respective second port **164***a-d* by an individual connector member.

[0048] With continued reference to FIG. **13**, the airflow control assembly **144** is fluidly connected to each of the air channels **156**. The airflow control assembly **144** is configured to selectively connect and/or selectively isolate one or more of the air channels **156**. The airflow control assembly **144** includes a valve body **174**. The valve body **174** houses a seal member **176** (see FIG. **14**). The seal member **176** is configured to move relative to the valve body **174**. As a nonlimiting example, the seal member **176** is configured to rotate relative to the valve body **174**. An indicator member **178** is coupled to the seal member **176**. The indicator member **178** is configured to rotate with the seal member **176**. An actuation assembly **180** is operatively connected to the seal member **176**. The actuation assembly **180** is configured to facilitate selective rotation of the seal member **176** relative to the valve body **174**. The actuation assembly **180** can include an actuation member **182**. As shown in FIG. **14**, in the illustrated embodiment, the actuation member **182** is a depressible button **182** that includes an angled projection **184**. The projection **184** is configured to selectively engage a cam member (not shown) coupled to the seal member **176**. By depressing the button **182** along a first axis A_1 towards the seal member **176**, the button **182** and angled projection **184** translate along the first axis A_1 into engagement with the cam member (not shown). This results in rotation of the seal member **176** relative to the valve body **174**. In other examples of embodiments, the actuation member **182** can be configured to rotate relative to the valve body **174** to facilitate rotation of the seal member **176**.

[0049] With reference to FIG. **15**, the valve body **174** defines a plurality of apertures **186**. Each aperture is associated with an associated air channel **156** to fluidly connect each air channel **156***a-d* to the valve body **174**.

[0050] The seal member **176** includes a plurality of seals that are arranged in different seal configurations. Movement of the seal member **176** relative to the valve body **174** is configured to align the seals of each seal configuration with the apertures **186**. In a first seal configuration (or a first position), the seals are arranged into a plurality of separate compartments. In the illustrated example of the multiport airflow control manifold assembly **100**, the seals are divided into four fluidly separate compartments. Each compartment is separated from an adjacent compartment by a circumferential radial divider. The first seal configuration is also separated from an adjacent seal configuration by a pair of elongated radial dividers extending along a length of the seal member. As such, the elongated radial dividers separate each seal configuration. In the first seal configuration, each compartment is isolated from an adjacent compartment. In turn, air cannot flow between compartments. When oriented within the valve body **174** into fluid communication with the air channels **156***a-d*, the first seal configuration results in each of the fluid conduits **54**, **58**, **62**, **66** and associated inflation zones **30**, **34**, **38**, **42** being fluidly isolated from each other. As such, air cannot travel between the inflation zones **30**, **34**, **38**, **42** via the seal member **176** in response to the first seal configuration.

[0051] In a second seal configuration (or second position), the seals are divided into a single compartment. The second seal configuration is separated from an adjacent seal configuration by a pair of elongated radial dividers extending along the length of the seal member **176**. In the second seal configuration, the single compartment facilitates a flow of air within the compartment. When oriented within the valve body **174** into fluid communication with the air channels **156***a-d*, the second seal configuration results in each of the fluid conduits **54**, **58**, **62**, **66** and associated inflation zones **30**, **34**, **38**, **42** being fluidly connected to each other. As such, air is free to travel between the inflation zones **30**, **34**, **38**, **42** via the seal member **176** in response to the second seal configuration.

[0052] In a third seal configuration (or a third position), the seals are divided into a pair of separated compartments. More specifically, the seals are divided into two fluidly separate compartments. The compartments are separated from the adjacent compartment by a single circumferential radial divider. The third seal configuration is also separated from an adjacent seal configuration by a pair of elongated radial dividers extending along a length of the seal member **176**. As such, the elongated radial dividers separate each seal configuration. In the third seal configuration, each of the two compartments are fluidly isolated. Thus, air cannot flow between the compartments. When oriented within the valve body **174** into fluid communication with the air channels **156***a-d*, the third seal configuration results in two fluid conduits being fluidly connected. Stated another way, the third seal configuration results in a first group of inflation zones **30**, **38** and a second group of inflation zones **34**, **42**.

The first group of inflation zones, which in the present embodiment is a first pair of inflation zones **30**, **38**, are fluidly connected. The second group of inflation zones, which in the present embodiment is a second pair of inflation zones **34**, **42**, are also fluidly connected. Air is configured to flow between the first pair of inflation zones **30**, **38**. Similarly, air is configured to flow between the second pair of inflation zones **34**, **42**. However, air is restricted from flowing from the first pair of inflation zones to the second pair of inflation zones, and vice versa. In the illustrated embodiment, the first pair of inflation zones correspond to the first side S_1, and the second pair of inflation zones correspond to the second side S_2 of the cellular cushion **10**. It should be appreciated that in other examples of embodiments, the third seal configuration can be reoriented with the associated fluid conduits **54**, **58**, **62**, **66** such that the third seal configuration results in the first pair of inflation zones corresponding to the front F inflation zones **30**, **34**, and the second pair of inflation zones correspond to the rear R inflation zones **38**, **42**. It should also be appreciated that the first group of inflation zones can be referred to as a first group of connectors, and the second group of inflation zones can be referred to as a second group of connectors. It should also be appreciated that while the first group of connectors are fluidly connected, and the second group of connectors are fluidly connected, the first and second group of connectors are fluidly isolated from each other.

[0053] As the seal member **176** moves relative to the valve body **174** between seal configurations, the indicator member **178** moves with the seal member **176**. As a nonlimiting example, as the seal member **176** rotates relative to the valve body **174** between seal configurations, the indicator member **178** rotates with the seal member **176**. The indicator member **178** includes a plurality of indicia, with each indicia associated with the selected seal configuration. For example, the indicia can be a respective identification, color, symbol, or any other suitable information to convey to a user the selected seal configuration.

[0054] It should be appreciated that the airflow control assembly **144** includes certain components and operates in a manner disclosed in U.S. Pat. No. 11,801,175, which is entitled "Multi-Position Airflow Control Assembly for an Air Cushion" and assigned to Permobil, Inc., the entire contents of which is herein incorporated by reference in its entirety.

[0055] With reference now to FIGS. **16-17**, the first portion **202** and a second portion **220** of the air supply assembly **200** are illustrated. The first portion **202** includes the plurality of air supply lines **204***a-d*. The second portion **220** also includes a plurality of air supply lines **204***a-d*. The first and second portions **202**, **220** are configured to selectively fasten together by a coupling assembly **222**. The coupling assembly includes a first coupling member **208** (shown as the male coupling member **208**) and a second coupling member **224** (shown as a female coupling member **224**). The male coupling member **208** includes a plurality of first ports, with each port associated with one of the air supply lines **204** of the first portion **202**. To block air loss in a disconnected configuration, each port of the male coupling member **208** has a valve that is biased to a closed position. The female coupling member **224** includes a plurality of second ports, with each port also associated with one of the air supply lines **204** of the second portion **220**. In response to the female coupling member **224** being positioned into engagement with the male coupling member **208**, the male coupling member **208** is configured to be received by the female coupling member **224**. More specifically, each first port of the coupling member **208** is configured to receive an associated second port of the coupling member **224**. Once received, each second port laterally slides the valve associated with each first port, overcoming the applied bias and fluidly connecting the air supply lines **204** of each portion **202**, **220**. The coupling members **208**, **224** are further configured to removably fasten together to facilitate the fluid connection. The coupling member **208**, **224** can also be selectively disengaged. Once disengaged (or unfastened) the male coupling member **208** is removed from the female coupling member **224**. In response to disengagement of the second ports from the first ports, the associated bias is reapplied to the valve of each first port, closing the first ports to limit air loss. It should be appreciated that in other examples of embodiments, the first portion **202** can include the female coupling member **224**, while the second portion **220** can include the male coupling member **208**. Stated another way, the male and female coupling members **208**, **224** can be associated with either the first or second portions **202**, **220**. In addition, it should be appreciated that the coupling assembly **222** can include any suitable system for selectively attaching and detaching the air supply lines **204** to facilitate air flow in a connected configuration and restrict airflow from escaping the cellular cushion **10** in a disconnected configuration.

[0056] FIG. **18** is a schematic diagram of an example of an embodiment of a remote air source assembly **300** (also referred to as an air source assembly **300**). The remote air source assembly **300** is configured to selectively connect to the multiport airflow control manifold assembly **100**, and in turn the cellular cushion **10**, by the air supply assembly **200**. FIG. **18** illustrates the second portion **220** of the air support assembly **200** (shown in FIGS. **16-17**).

[0057] The remote air source assembly **300** includes a pump **304** (also referred to as an air pump **304**) that is operably and fluidly connected to a valve array **306**. The valve array **306** includes a plurality of valves **308**, which are each illustrated as solenoid valves **308**. The valves **308** of the valve array **306** are configured to selectively open to facilitate air flow from the pump **304** through the valve **308** to the associated air supply line **204** to facilitate inflation of each respective inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**. The valves **308** of the valve array **306** can also be configured to selectively open to facilitate air flow from each associated air supply line **204** to atmosphere to facilitate deflation of each respective inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**. A pressure sensor **312** can also be positioned downstream of each valve **308** to detect a pressure in the associated inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**. The remote air source assembly **300** can also include a controller (not shown) to facilitate operation of the pump **304** and valve array **306**, and to receive detected pressure readings from each pressure sensor **312**. The remote air source assembly **300** can further include a power source (not shown) such as a rechargeable battery or other suitable power source to operate the pump **304**, valve array **306**, pressure sensors **312**, controller, and other associated components.

[0058] In operation, the multiport airflow control manifold assembly **100** is configured to facilitate inflation and/or deflation of the associated inflation zones **30**, **34**, **38**, **42** of the cellular cushion **10**. To facilitate inflation, the pump **304** of the remote air source assembly **300** can operate. Depending on the inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10** that requires inflation, the associated valve **308** of the valve array **306** can be opened to facilitate airflow from the remote air source assembly **300** to the associated inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**.

[0059] For example, to facilitate inflation of the first inflation zone **30**, the pump **304** is operational and a first valve **308***a* actuates to an open configuration. Air flow from the pump **304** travels through the first valve **308***a*, exiting the remote air source assembly **300** and traveling through the first air supply line **204***a*. The air flow then enters the first second port **164***a* of the manifold assembly **140**. The air flow travels from the first second port **164***a* to the first air channel **156***a*, where the air flow exits the manifold assembly **140** through the first first port **160***a*. The air flow is then directed to the first inflation zone **30** through the first fluid conduit **54** of the cellular cushion **10**. Once a desired inflation level is achieved, the remote air source assembly **300** closes the associated first valve **308***a* to terminate air flow to the first inflation zone **30**.

[0060] To facilitate inflation of the second inflation zone **34**, the pump **304** is operational and a second valve **308***b* actuates to an open configuration. Air flow from the pump **304** travels through the second valve **308***b*, exiting the remote air source assembly **300** and traveling through the second air supply line **204***b*. The air flow then enters the second second port **164***b* of the manifold assembly **140**. The air flow travels from the second second port **164***b* to the second air channel **156***b*, where the air flow exits the manifold assembly **140** through the second first port **160***b*. The air flow is then directed to the second inflation zone **34** through the second fluid conduit **58** of the cellular cushion **10**. Once a desired inflation level is achieved, the remote air source assembly **300** closes the associated second valve **308***b* to terminate air flow to the second inflation zone **34**.

[0061] To facilitate inflation of the third inflation zone **38**, the pump **304** is operational and a third valve **308***c* actuates to an open configuration. Air flow from the pump **304** travels through the third valve **308***c*, exiting the remote air source assembly **300** and traveling through the third air supply line **204***c*. The air flow then enters the third second port **164***c* of the manifold assembly **140**. The air flow travels from the third second port **164***c* to the third air channel **156***c*, where the air flow exits the manifold assembly **140** through the third first port **160***c*. The air flow is then directed to the third inflation zone **38** through the third fluid conduit **62** of the cellular cushion **10**. Once a desired inflation level is achieved, the remote air source assembly **300** closes the associated third valve **308***c* to terminate air flow to the third inflation zone **38**.

[0062] To facilitate inflation of the fourth inflation zone **42**, the pump **304** is operational and a fourth valve **308***d* actuates to an open configuration. Air flow from the pump **304** travels through the fourth valve **308***d*, exiting the remote air source assembly **300** and traveling through the fourth air supply line **204***d*. The air flow then enters the fourth second port **164***d* of the manifold assembly **140**. The air flow travels from the fourth second port **164***d* to the fourth air channel **156***d*, where the air flow exits the manifold assembly **140** through the fourth first port **160***d*. The air flow is then directed to the fourth inflation zone **42** through the fourth fluid conduit **62** of the cellular cushion **10**. Once a desired inflation level is achieved, the remote air source assembly **300** closes the associated fourth valve **308***d* to terminate air flow to the fourth inflation zone **42**.

[0063] Once each inflation zone **30**, **34**, **38**, **42** is inflated to a desired inflation level, the multiport airflow control manifold assembly **100** can facilitate maintaining of a desired (or target) inflation level. For example, each pressure sensor **312***a*, **312***b*, **312***c*, **312***d* is configured to measure a pressure in an associated inflation zone **30**, **34**, **38**, **42** of the cellular cushion **10**. For example, a first pressure sensor **312***a* detects a pressure in the first inflation zone **30**, a second pressure sensor **312***b* detects a pressure in the second inflation zone **34**, a third pressure sensor **312***c* detects a pressure in the third inflation zone **38**, and a fourth pressure sensor **312***d* detects a pressure in the fourth inflation zone **42**. In response to a detected pressure dropping below a desired pressure, the remote air source assembly **300** can initiate operation of the pump **304** and then open the associated valve **308***a-d* to reinflate the associated inflation zone **30**, **34**, **38**, **42**. Once a desired pressure is achieved, the remote air source assembly **300** can close the associated valve **308***a-d*, and when all valves **308***a-d* are closed, can terminate operation of the pump **304**.

[0064] Should an inflation zone **30**, **34**, **38**, **42** be overinflated, or a target pressure reduced for one or more inflation zones **30**, **34**, **38**, **42**, the remote air source assembly **300** can also facilitate deflation of one or more inflation zones **30**, **34**, **38**, **42**. For example, in response to a detected pressure being above a desired pressure, or a target desired pressure being reduced, for one or more inflation zones **30**, **34**, **38**, **42**, the remote air source assembly **300** can open the associated valve **308***a-d* to discharge to atmosphere. Air from the associated inflation zone **30**, **34**, **38**, **42** can travel from the associated inflation zone **30**, **34**, **38**, **42**, through the associated fluid conduit **54**, **58**, **62**, **66**, into the associated air channel **156***a-d* from the associated first port **160***a-d*, out of the associated air channel **156***a-d* from the associated second port **164***a-d*, through the associated air supply line **204***a-d*, and then through the associated valve **308***a-d* to be discharged to atmosphere. Once an inflation zone **30**, **34**, **38**, **42** is deflated to a desired pressure, the remote air source assembly **300** can close the associated valve **308***a-d*. It should be appreciated that alternatively, a user can deflate the inflation zones **30**, **34**, **38**, **42** through the air valve **70**. Once an amount of air is discharged, the inflation zones **30**, **34**, **38**, **42** can reach a level of equilibrium, the remote air source assembly **300** can measure and control the inflation level of the inflation zones **30**, **34**, **38**, **42** relative to the desired pressure.

[0065] In addition to being configured to facilitate inflation and/or deflation of the associated inflation zones **30**, **34**, **38**, **42** of the cellular cushion **10**, the multiport airflow control manifold assembly **100** is configured to selectively connect and/or selectively isolate one or more of the inflation zones **30**, **34**, **38**, **42**. For example, actuation of the airflow control assembly **144** to the first seal configuration results in fluidly isolating all of the inflation zones **30**, **34**, **38**, **42**. Air cannot travel within the seal member **176** between air channels **156***a-d*. Stated another way, each air channel **156***a-d* is fluidly isolated from all of the other air channels **156***a-d*. Air can only travel through each respective air channel **156***a-d* of the manifold assembly, entering and exiting the associated first and second ports **160***a-d*, **164***a-d*.

[0066] Actuation of the airflow control assembly **144** to the second seal configuration results in fluidly connecting all of the inflation zones **30**, **34**, **38**, **42**. Air can travel within the seal member **176** between all of the air channels **156***a-d*. Stated another way, all of the air channels **156***a-d* are fluidly connected to each other by the seal member **176**. Air can not only travel through each respective air channel **156***a-d* of the manifold assembly, entering and exiting the associated first and second ports **160***a-d*, **164***a-d*, but air also connects all of the air channels **156***a-d* together within the seal member **176**.

[0067] Actuation of the airflow control assembly **144** to the third seal configuration results in fluidly connecting groups of inflation zones **30**, **34**, **38**, **42**. In one example, in the third seal configuration, the seal member connects the first group of inflation zones **30**, **38** and the second group of inflation zones **34**, **42**. In this configuration, the first and third air channels **156***a*, **156***c* are fluidly connected within the seal member **176**. As such, air can travel between the first and third air channels **156***a*, **156***c* in the seal member **176**. In addition, the second and fourth air channels **156***b*, **156***d* are fluidly connected within the seal member **176**. As such, air can travel between the second and fourth air channels **156***b*, **156***d* in the seal member **176**. Stated another way, air in the first group of inflation zones **30**, **38** can enter and exit through each air channel **156***a*, **156***c* through the associated first ports **160***a*, **160***c* and second ports **164***a*, **164***c*, and further air can travel between the first and third air channels **156***a*, **156***c* within the seal member **176**. Air in the second group of inflation zones **34**, **42** can enter and exit through each air channel **156***b*, **156***d* through the associated first ports **160***b*, **160***d* and second ports **164***b*, **164***d*, and further air can travel between the second and fourth air channels **156***b*, **156***d* within the seal member **176**.

[0068] In another example, in the third seal configuration, the seal member connects the first group of inflation zones **30**, **34** and the second group of inflation zones **38**, **42**. In this configuration, the first and second air channels **156***a*, **156***b* are fluidly connected within the seal member **176**. As such, air can travel between the first and second air channels **156***a*, **156***b* in the seal member **176**. In addition, the third and fourth air channels **156***c*, **156***d* are fluidly connected within the seal member **176**. As such, air can travel between the third and fourth air channels **156***c*, **156***d* in the seal member **176**. Stated another way, air in the first group of inflation zones **30**, **34** can enter and exit through each air channel **156***a*, **156***b* through the associated first ports **160***a*, **160***b* and second ports **164***a*, **164***b*, and further air can travel between the first and second air channels **156***a*, **156***b* within the seal member **176**. Air in the second group of inflation zones **38**, **42** can enter and exit through each air channel **156***c*, **156***d* through the associated first ports **160***c*, **160***d* and second ports **164***c*, **164***d*, and further air can travel between the third and fourth air channels **156***c*, **156***d* within the seal member **176**. It should be appreciated that in some examples of embodiments, the alternative third seal configuration (grouping zones **30**, **34** and **38**, **42**) can be a fourth seal configuration. It should also be appreciated that any suitable grouping of one or more zones can be achieved, and further the zones can be grouped in uneven numbers. For example, in an example of an embodiment of a cushion with three zones, a first group can include one zone **30** (or a pair of zones **30**, **34**) and a second group can include a pair of zones **34**, **38** (or one zone **38**).

[0069] FIGS. **1-18** disclose an embodiment of a cellular cushion **10** that includes four inflation zones **30**, **34**, **38**, **42**. Accordingly, the associated multiport airflow control manifold assembly **100** includes four air channels **156***a-d*, the air supply assembly **200** includes four air supply lines **204**, and the remote air source assembly **300** includes a valve array **306** that includes four valves **308**. It should be appreciated that this is one example of an embodiment, and the multiport airflow control manifold assembly **100** and associated components can easily be adapted for use with a cellular cushion **10** having fewer or more than fourth inflation zones **30**, **34**, **38**, **42**. FIGS. **19-21** illustrate non-limiting alternative examples. It should be appreciated that FIGS. **19-21** use similar identification numbers to identify similar components.

[0070] With reference to FIG. **19**, an embodiment of the cellular cushion **10***a* is illustrated having two inflation zones **30**, **34**. Since this embodiment of the cellular cushion **10***a* includes only two inflation zones **30**, **34**, the multiport airflow control manifold assembly **100***a* is modified to include two air channels **156** (not shown). As such, the air supply assembly **200***a* is modified to have two air supply lines **204***a*, **204***b* fluidly connecting the remote air source assembly **300***a* to the multiport airflow control manifold assembly **100***a*. In addition, the first and second coupling members **208***a*, **224***a* are modified to each have two ports to facilitate connection of the portions of the two air supply lines **204***a*, **204***b*. The remote air source assembly **300***a* also includes a valve array **306** (not shown) that includes two valves **308** (not shown), one associated with each air supply line **204***a*, **204***b*. The components otherwise operate as described in association with the cellular cushion **10** and the multiport airflow control manifold assembly **100**.

[0071] With reference to FIG. **20**, an embodiment of the cellular cushion **10***b* is illustrated having three inflation zones **30**, **34**, **38**. Since this embodiment of the cellular cushion **10***b* includes only three inflation zones **30**, **34**, **38**, the multiport airflow control manifold assembly **100***b* is modified to include three air channels **156** (not shown). As such, the air supply assembly **200***b* is modified to have three air supply lines **204***a*, **204***b*, **204***c* fluidly connecting the remote air source assembly **300***b* to the multiport airflow control manifold assembly **100***b*. In addition, the first and second coupling members **208***b*, **224***b* are modified to each have three ports to facilitate connection of the portions of the three air supply lines **204***a*, **204***b*, **204***c*. The remote air source assembly **300***b* also includes a valve array **306** (not shown) that includes three valves **308** (not shown), one associated with each air supply line **204***a*, **204***b*, **204***c*. The components otherwise operate as described in association with the cellular cushion **10** and the multiport airflow control manifold assembly **100**.

[0072] With reference to FIG. **20**, an embodiment of the cellular cushion **10***c* is illustrated having a single inflation zone **30**. Since this embodiment of the cellular cushion **10***c* includes only one inflation zone **30**, the multiport airflow control manifold assembly **100***c* is modified to include one air channel **156** (not shown). In addition, the multiport airflow control manifold assembly **100***c* is arranged to eliminate the airflow control assembly **144**. The air supply assembly **200***c* is modified to have one air supply line **204***a* fluidly connecting the remote air source assembly **300***c* to the multiport airflow control manifold assembly **100***c*. In addition, the first and second coupling members **208***c*, **224***c* are modified to each have one port to facilitate connection of the portions of the air supply line **204***a*. The remote air source assembly **300***c* also includes a valve array **306** (not shown) that includes one valve **308** (not shown) associated with the air supply line **204***a*. The components otherwise operate as described in association with the cellular cushion **10** and the multiport airflow control manifold assembly **100**.

[0073] One or more aspects of the multiport airflow control manifold assembly **100** provides certain advantages outlined above. These and other advantages are realized by the disclosure provided herein.

What is claimed is:

1. A multiport airflow control manifold assembly comprising:

a valve assembly including:

a manifold assembly defining a first air channel, a second air channel, and a third air channel;

a first first port fluidly connected to first air channel;

a first second port fluidly connected to the first air channel;

a second first port fluidly connected to second air channel;

a second second port fluidly connected to the second air channel;

a third first port fluidly connected to third air channel;

a third second port fluidly connected to the third air channel; and

an airflow control assembly fluidly connected to the manifold assembly, the airflow control assembly including a seal member configured to move between a first seal configuration and a second seal configuration,

wherein in the first seal configuration, the seal member is configured to fluidly isolate the first air channel, the second air channel, and the third air channel, and

wherein in the second seal configuration, the seal member is configured to fluidly connect the first air channel, the second air channel, and the third air channel.

2. The multiport airflow control manifold assembly of claim **1**, wherein the first second port is fluidly connected to the first air channel downstream of the seal member and upstream of the first first port.

3. The multiport airflow control manifold assembly of claim **1**, wherein the second second port is fluidly connected to the second air channel downstream of the seal member and upstream of the second first port.

4. The multiport airflow control manifold assembly of claim **1**, wherein the third second port is fluidly connected to the third air channel downstream of the seal member and upstream of the third first port.

5. The multiport airflow control manifold assembly of claim **1**, wherein the seal member is configured to move to a third seal configuration, wherein in the third seal configuration, the seal member is configured to fluidly isolate the first air channel, and fluidly connect the second air channel and the third air channel.

6. The multiport airflow control manifold assembly of claim **1**, wherein the first second port, the second second port, and the third second port are configured to be fluidly connected to an air source assembly, the air source assembly is configured to provide a flow of air to the manifold assembly.

7. The multiport airflow control manifold assembly of claim **6**, wherein the first first port is configured to be fluidly connected to a first inflation zone of a cellular cushion, the second first port is configured to be fluidly connected to a second inflation zone of the cellular cushion, and the third first port is configured to be fluidly connected to a third inflation zone of the cellular cushion.

8. The multiport airflow control manifold assembly of claim **6**, wherein the air source assembly includes a pump, the pump is configured to provide the flow of air.

9. The multiport airflow control manifold assembly of claim **6**, wherein the air source assembly is fluidly connected to the first second port, the second second port, and the third second port by an air supply assembly.

10. The multiport airflow control manifold assembly of claim **9**, wherein the air supply assembly includes a first air supply line fluidly connecting the air source assembly to the first second port, a second air supply line fluidly connecting the air source assembly to the second second port, and a third air supply line fluidly connecting the air source assembly to the third second port.

11. The multiport airflow control manifold assembly of claim **10**, wherein the air supply assembly includes a coupling assembly defining a first coupling member and a second coupling member, the first coupling member configured to removably connect to the second coupling member to fluidly connect the first, second, and third air supply lines between the air supply assembly and the manifold assembly.

12. The multiport airflow control manifold assembly of claim **11**, wherein in response to disconnecting the first coupling member and the second coupling member, the first, second, and third air supply lines are fluidly disconnected between the air supply assembly and the manifold assembly.

13. The multiport airflow control manifold assembly of claim **1**, wherein the airflow control assembly includes a valve body, the seal member is received by the valve body.

14. The multiport airflow control manifold assembly of claim **13**, wherein the seal member is configured to rotate relative to the valve body between seal configurations.

15. The multiport airflow control manifold assembly of claim **13**, wherein the valve body defines a plurality of apertures, each aperture fluidly connects the first air channel, the second air channel, and the third air channel to the seal member.

16. A multiport airflow control manifold assembly configured to connect an air supply assembly to a cellular cushion, the manifold assembly comprising:

a manifold assembly defining a first air channel, a second air channel, and a third air channel;

a first first port and a first second port fluidly connected to the first air channel;

a second first port and a second second port fluidly connected to the second air channel;

a third first port and a third second port fluidly connected to the third air channel;

an airflow control assembly fluidly connected to the manifold assembly, the airflow control assembly including a valve body fluidly connected to the first air channel, the second air channel, and the third air channel, and a seal member configured to move relative to the valve body between a first position and a second position,

wherein in the first position, the seal member is configured to fluidly isolate the first air channel, the second air channel, and the third air channel, and wherein in the second position, the seal member is configured to fluidly isolate the first air channel, and fluidly connect the second air channel and the third air channel.

17. The multiport airflow control manifold assembly of claim **16**, wherein the first second port is fluidly connected to the first air channel upstream of the valve body and downstream of the first first port,

wherein the second second port is fluidly connected to the second air channel upstream of the valve body and downstream of the second first port, and

wherein the third second port is fluidly connected to the third air channel upstream of the valve body and downstream of the third first port.

18. The multiport airflow control manifold assembly of claim **16**, wherein the first second port, the second second port, and the third second port are each configured to be fluidly connected to the air supply assembly.

19. The multiport airflow control manifold assembly of claim **16**, wherein the first first port is configured to be fluidly connected to a first inflation zone of the cellular cushion, the second first port is configured to be fluidly connected to a second inflation zone of the cellular cushion, and the third first port is configured to be fluidly connected to a third inflation zone of the cellular cushion.

20. The multiport airflow control manifold assembly of claim **16**, wherein the seal member is configured to move to a third position, wherein in the third position the seal member fluidly connects the first air channel, the second air channel, and the third air channel.

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